Nametable Mapping

File: nametable_mapping.kicad_sch

CPU Address Bus

File: cpuAddressBus.kicad_sch

Register Access

File: register_access.kicad_sch

PRG Banking

File: prg_banking.kicad_sch

PPU Address Bus

File: ppu_address_bus.kicad_sch

Reset Detection

File: reset_detection.kicad_sch

CPU Data Bus

File: cpu_data_bus.kicad_sch

PCM DAC

File: pcm_dac.kicad_sch

Scanline Detection

File: scanline_detection.kicad_sch

V-Split

File: v_split.kicad_sch

PPU Fetch Counting

File: ppu_fetch_counting.kicad_sch

CHR A0,A1,A2

File: chr_a0_a1_a2.kicad_sch

ExRAM Mode

File: exram_mode.kicad_sch

Fill Mode

File: fill_mode.kicad_sch

Pulse Length Lookup Table

File: pulse_length_lookup_table.kicad_sch

Pulse Timer

File: pulse_timer.kicad_sch

Pulse Duty Envelope Volume

File: pulse_duty_envelope_volume.kicad_sch

Pulse DAC

File: Pulse DAC.kicad_sch

PPU Data Bus

File: ppu_data_bus.kicad_sch

CHR Bank Registers

File: chr_bank_registers.kicad_sch

CHR Bank Selection Logic

File: chr_bank_selection_logic.kicad_sch

CHR Address Bus 10 to 14

File: chr_address_bus_10_to_14.kicad_sch

CHR Address Bus 15 to 19

File: chr_address_bus_15_to_19.kicad_sch

Multiplier

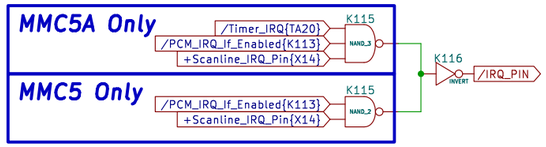
File: multiplier.kicad_sch

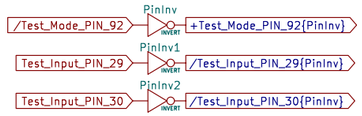
MMC5A Timer

File: mmc5a_timer.kicad_sch

Artifacts

File: artifacts.kicad_sch



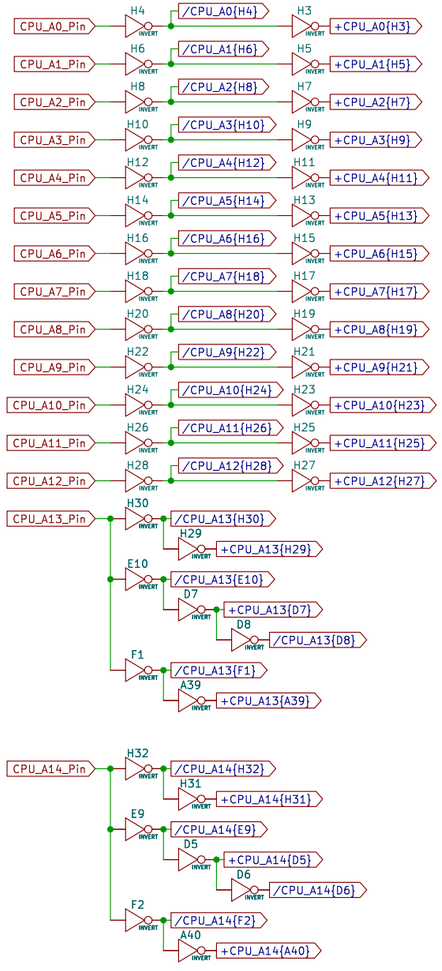


Sheet: /
File: MMC5.kicad_sch

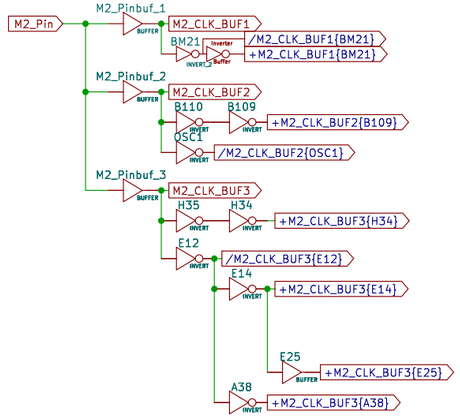
Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 1/27

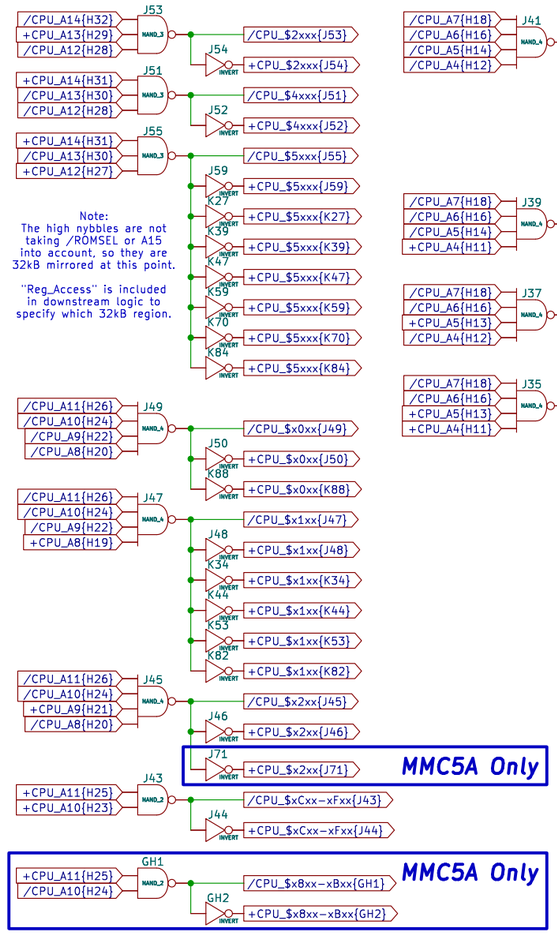
CPU Address Bus Buffering



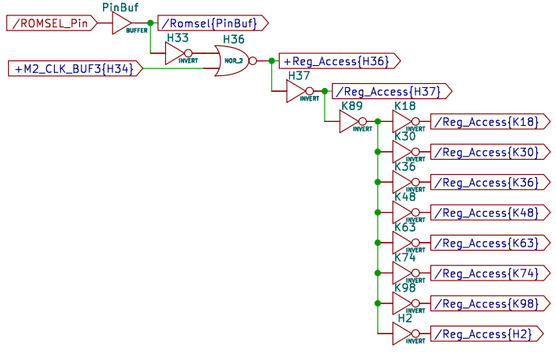
M2 Clock Buffering



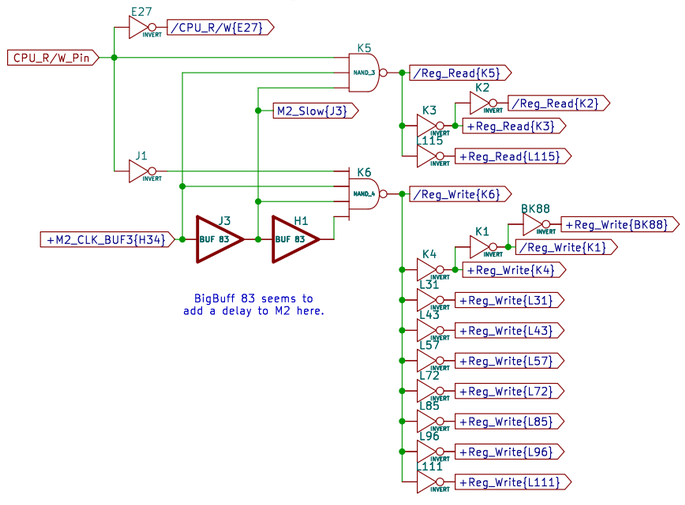
CPU Address Bus Nybble Decoding



/ROMSEL, Register Access



CPU R/W, Register Read/Write



Sheet: /CPU Address Bus/
File: cpuAddressBus.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**

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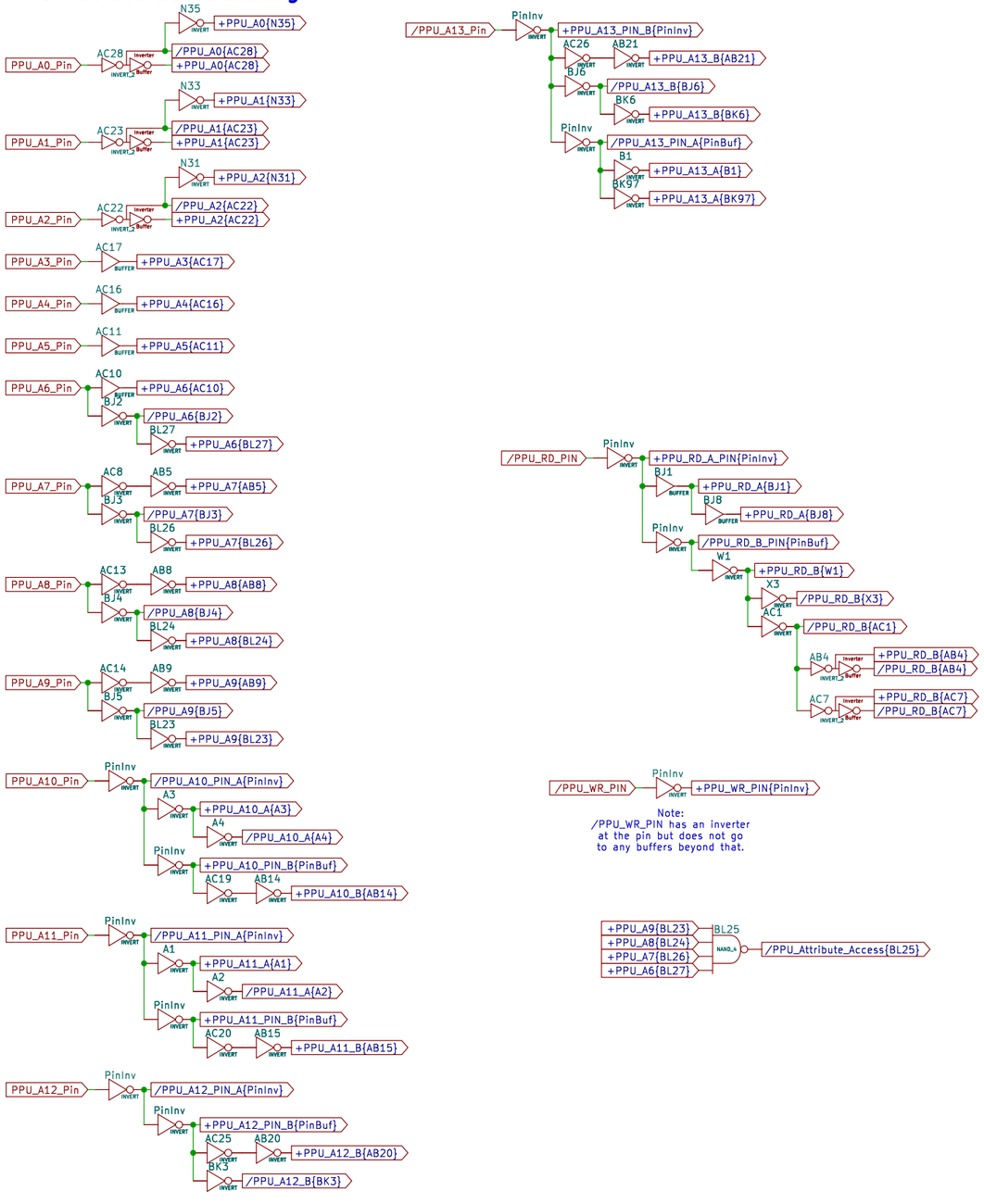
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L106
NOR_2
+Reg_Write{L111}
L105
NAND_2
+CPU_$2000_Access{L106}
/CPU_$2000_WR{L105}
+CPU_$2xxx{J54}
+CPU_$x0xx{K88}
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+CPU_$xxx1{J32}
K91
NAND_4
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L108
NOR_2
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L107
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L91
NOR_2
+Reg_Write{L85}
+Reg_Read{K3}
L90
NAND_2
/CPU_$5205_RD{L90}
L89
NAND_2
+CPU_$5205_Access{L91}
/CPU_$5205_WR{L89}
+CPU_$5xxx{K70}
+CPU_$x2xx{J46}
+CPU_$xx0x{K68}
+CPU_$xxx6{J22}
K69
NAND_4
/CPU_$5206{K69}
/Reg_Access{K74}
L92
NOR_2
+Reg_Write{L85}
+Reg_Read{K3}
L94
NAND_2
/CPU_$5206_RD{L94}
L93
NAND_2
+CPU_$5206_Access{L92}
/CPU_$5206_WR{L93}
+CPU_$5xxx{K84}
+CPU_$xCxx-xFxx{J44}
K95
NAND_2
/Reg_Access{K98}
L117
NOR_2
+CPU_ExRAM_Access{L117}
MMC5A Only
+CPU_$5xxx{K84}
+CPU_$x8xx-xBxx{GH2}
J75
NAND_2
/Reg_Access{K98}
J76
NOR_2
+CPU_5800-5BFF_Access{J76}
+CPU_A1{H5}
/CPU_A2{H8}
+CPU_A3{H9}
+CPU_A4{H11}
+CPU_A5{H13}
+CPU_A6{H15}
+CPU_A7{H17}
J4
NAND_7
+CPU_A8{H19}
+CPU_A9{H21}
+CPU_A10{H23}
+CPU_A11{H25}
+CPU_A12{H27}
+CPU_A13{H29}
+CPU_A14{H31}
J5
NAND_7
J6
NOR_2
+Reg_Access{H36}
J7
INVERT
J8
NOR_2
+Reg_Read{K3}
+CPU_$FFFA/B_Access{J8}
L21
NAND_2
/CPU_$FFFA/B_RD{L21}
/CPU_$5120_WR{L56}
/CPU_$5121_WR{L58}
/CPU_$5122_WR{L62}
/CPU_$5123_WR{L64}
/CPU_$5124_WR{L66}
/CPU_$5125_WR{L69}
/CPU_$5126_WR{L70}
/CPU_$5127_WR{L74}
AF1
NAND_8
AE1
INVERT
/CPU_$5120/1/2/3/4/5/6/7_WR{AE1}
/CPU_$5128_WR{L98}
/CPU_$5129_WR{L99}
/CPU_$512A_WR{L102}
/CPU_$512B_WR{L104}
AF2
NAND_4
AE2
INVERT
/CPU_$5128/9/A/B_WR{AE2}
MMC5A Only
+CPU_$2xxx{J54}
+CPU_$x0xx{J50}
+CPU_$xx0x{K90}
+CPU_$xxx6{J22}
J72
NAND_4
/CPU_$2006{J72}
/Reg_Access{K98}
J73
NOR_2
+Reg_Write{L111}
J74
NAND_2
+CPU_$2006_Access{J73}
/CPU_$2006_WR{J74}
+CPU_$5xxx{J59}
+CPU_$x2xx{J46}
+CPU_$xx0x{J42}
+CPU_$xxx7{J20}
L142
NAND_4
/CPU_$5207{L142}
/Reg_Access{K18}
L143
NOR_2
+Reg_Write{K4}
L144
NAND_2
+CPU_$5207_Access{L143}
/CPU_$5207_WR{L144}
+CPU_$5xxx{K59}
+CPU_$x2xx{J71}
+CPU_$xx0x{K90}
+CPU_$xxxA{J12}
K121
NAND_4
/CPU_$520A{K121}
/Reg_Access{H2}
K122
NOR_2
+Reg_Write{L96}
K123
NAND_2
+CPU_$520A_Access{K122}
/CPU_$520A_WR{K123}
+CPU_$5xxx{K39}
+CPU_$x2xx{J46}
+CPU_$xx0x{J42}
+CPU_$xxx8{J18}
L145
NAND_4
/CPU_$5208{L145}
/Reg_Access{K30}
L146
NOR_2
+Reg_Write{L31}
+Reg_Read{K3}
L148
NAND_2
/CPU_$5208_RD{L148}
L147
NAND_2
+CPU_$5208_Access{L146}
/CPU_$5208_WR{L147}
+CPU_$5xxx{K59}
+CPU_$x2xx{J71}
+CPU_$xx0x{K90}
+CPU_$xxx9{J16}
K117
NAND_4
/CPU_$5209{K117}
/Reg_Access{H2}
K118
NOR_2
+Reg_Write{L96}
+Reg_Read{L115}
K120
NAND_2
/CPU_$5209_RD{K120}
K119
NAND_2
+CPU_$5209_Access{K118}
/CPU_$5209_WR{K119}

Sheet: /Register Access/
File: register_access.kicad_sch

Title:

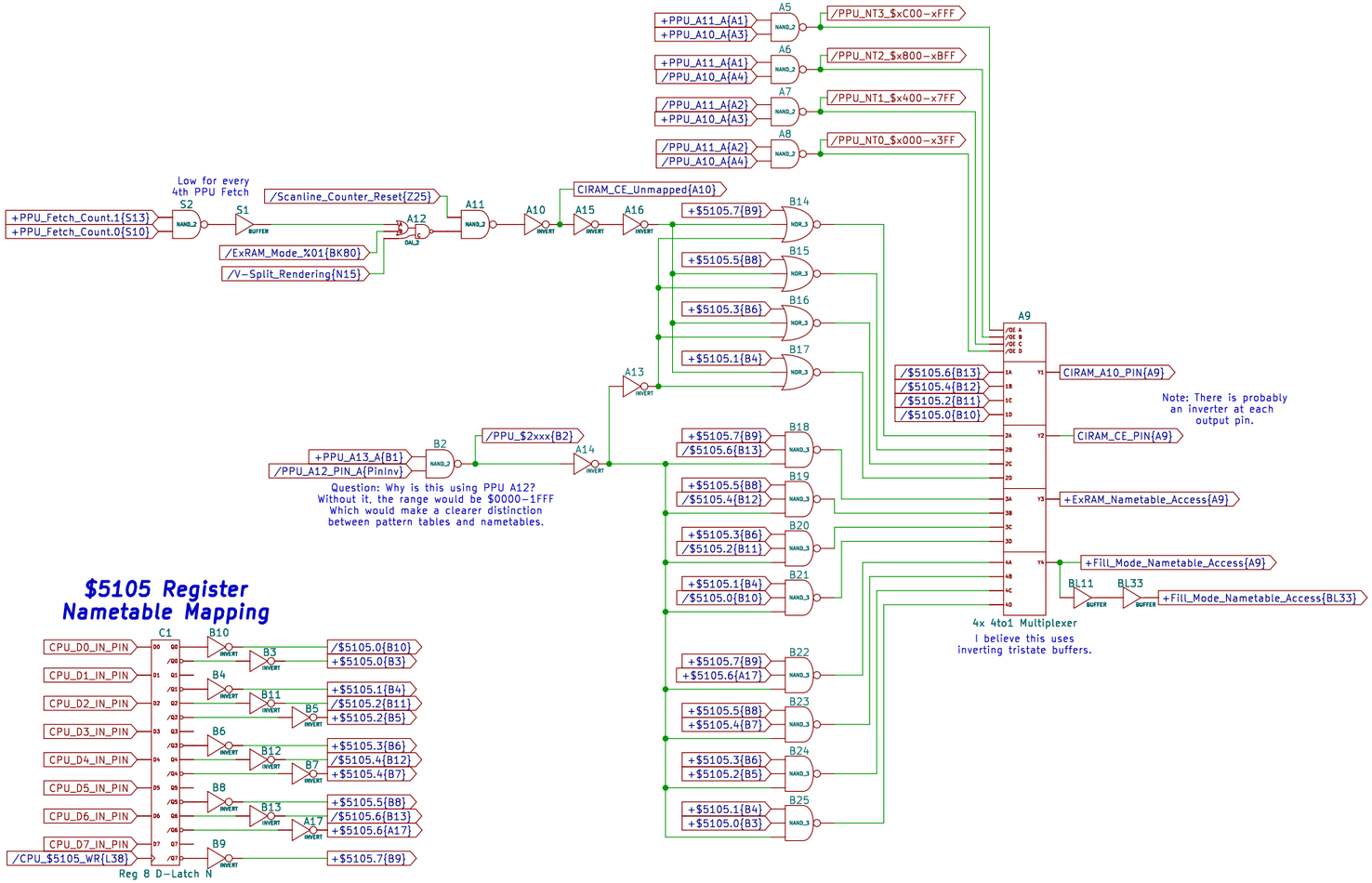
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 3/27

PPU Address Bus Buffering



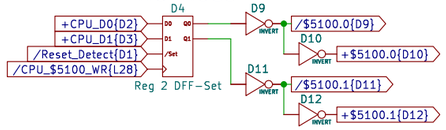
Sheet: /PPU Address Bus/	
File: ppu_address_bus.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 4/27

$5105 Register Nametable Mapping

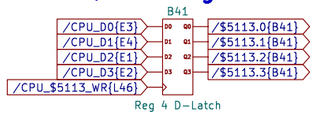


Sheet: /Nametable Mapping/		
File: nametable_mapping.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 5/27

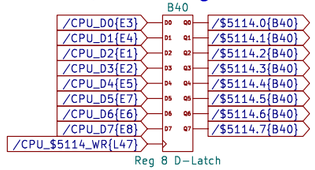
$5100 Register
PRG Mode



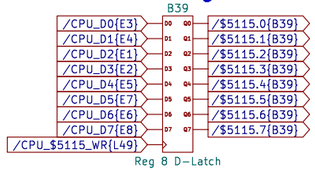
PRG Bank Registers
$5113 Register



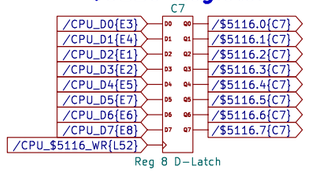
$5114 Register



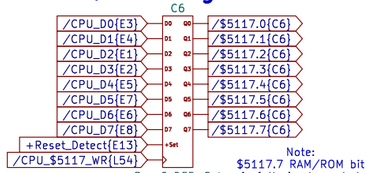
$5115 Register



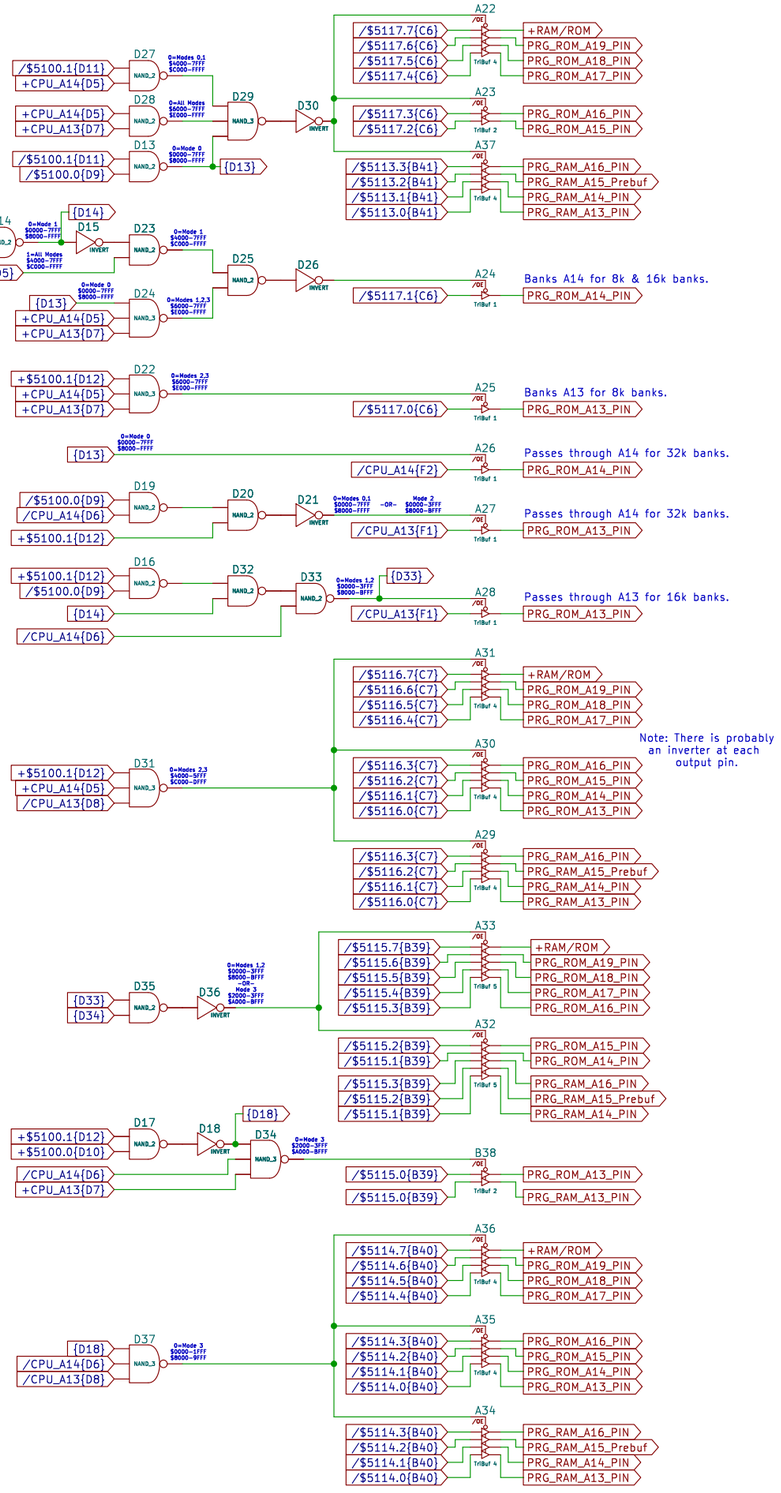
$5116 Register

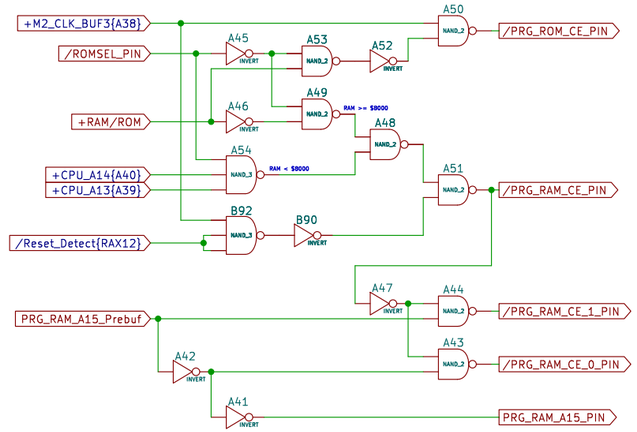


$5117 Register

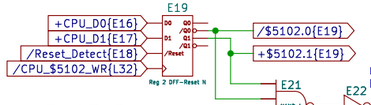


Note: $5117.7 RAM/ROM bit is fully implemented but is somehow being prevented from entering RAM mode when bench testing.

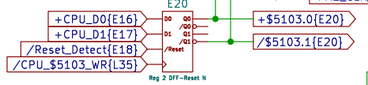




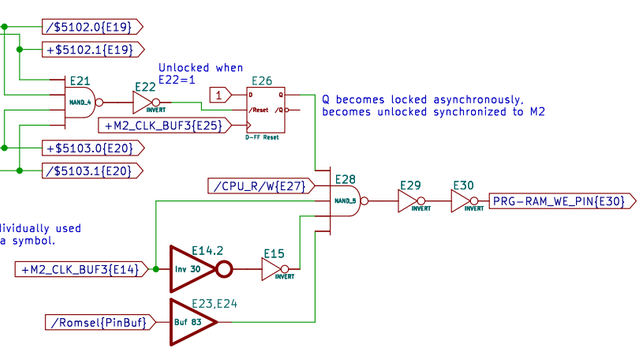
$5102 Register
PRG-RAM Write Protection



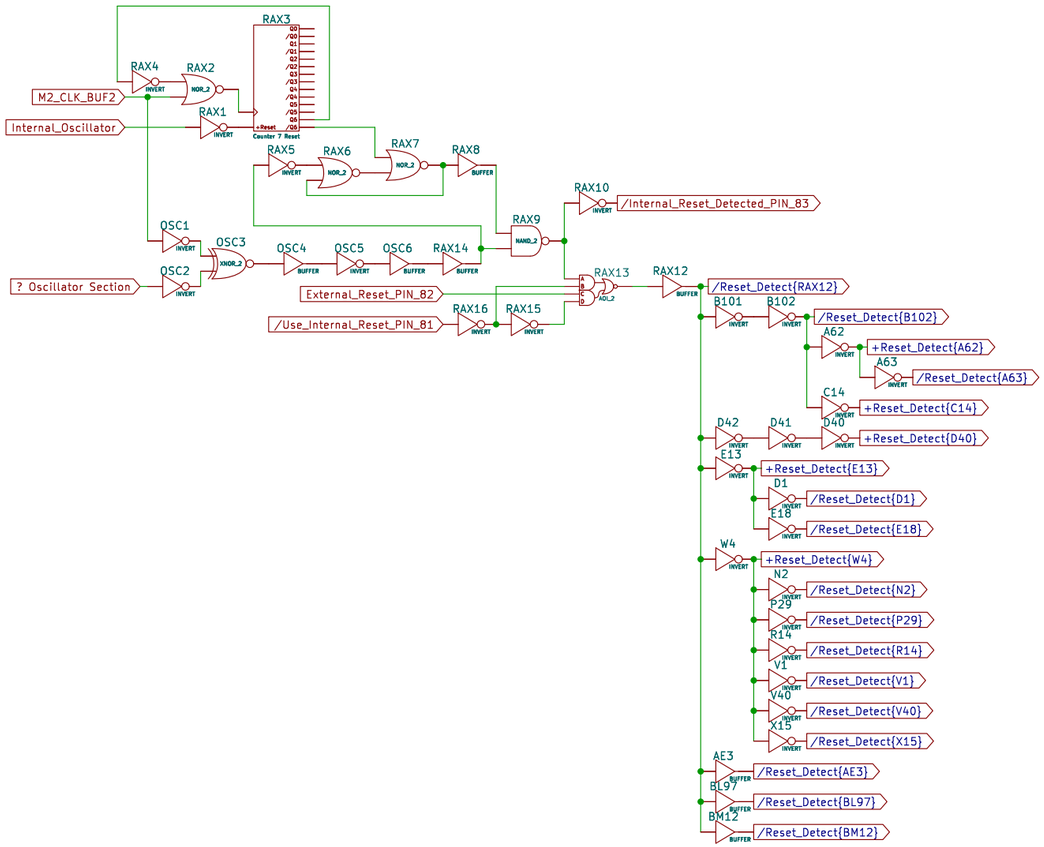
$5103 Register

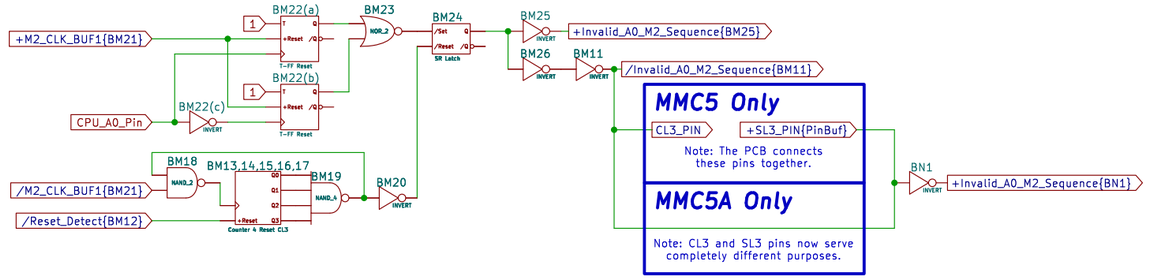


Note: /Set vs. /Reset inputs are individually used but not easily shown here with a symbol.



Sheet: /PRG Banking/
File: prg_banking.kicad_sch
Title:
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 6/27





Sheet: /Reset Detection/
File: reset_detection.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 7/27

Incoming CPU Data Bus Buffering Bonanza

CPU_D0_IN_PIN → E3 → /CPU_D0{E3}
- D2 → +CPU_D0{D2}
- E16 → +CPU_D0{E16}
- G15 → G16 → +CPU_D0{G16}
  - L122 → /CPU_D0{L122}
- X18 → /CPU_D0{X18}
  - P26 → +CPU_D0{P26}
  - Y4 → +CPU_D0{Y4}
- AE16 → /CPU_D0{AE16}
- AE18 → /CPU_D0{AE18}
- MUL19 → +CPU_D0{MUL19}
- TF1 → +CPU_D0{TF1}

MMC5A Only

CPU_D1_IN_PIN → E4 → /CPU_D1{E4}
- D3 → +CPU_D1{D3}
- E17 → +CPU_D1{E17}
- G13 → G14 → +CPU_D1{G14}
  - L123 → /CPU_D1{L123}
- X19 → /CPU_D1{X19}
  - P19 → +CPU_D1{P19}
  - Y5 → +CPU_D1{Y5}
- AE15 → /CPU_D1{AE15}
- AE19 → /CPU_D1{AE19}
- MUL20 → +CPU_D1{MUL20}
- TF2 → +CPU_D1{TF2}

MMC5A Only

CPU_D2_IN_PIN → E1 → /CPU_D2{E1}
- G11 → G12 → +CPU_D2{G12}
  - L124 → /CPU_D2{L124}
- X20 → /CPU_D2{X20}
  - P13 → +CPU_D2{P13}
  - Y11 → +CPU_D2{Y11}
- AE20 → /CPU_D2{AE20}
- MUL21 → +CPU_D2{MUL21}
- TF3 → +CPU_D2{TF3}

MMC5A Only

CPU_D3_IN_PIN → E2 → /CPU_D3{E2}
- G9 → G10 → +CPU_D3{G10}
  - E55 → /CPU_D3{E55}
  - E60 → +CPU_D3{E60}
  - L125 → /CPU_D3{L125}
- X21 → /CPU_D3{X21}
  - Y12 → +CPU_D3{Y12}
- AE25 → /CPU_D3{AE25}
- MUL22 → +CPU_D3{MUL22}
- TF4 → +CPU_D3{TF4}

MMC5A Only

CPU_D4_IN_PIN → E5 → /CPU_D4{E5}
- X22 → /CPU_D4{X22}
  - Y18 → +CPU_D4{Y18}
- G7 → G8 → +CPU_D4{G8}
  - E56 → /CPU_D4{E56}
  - E61 → +CPU_D4{E61}
  - L126 → /CPU_D4{L126}
- AE24 → /CPU_D4{AE24}
- MUL23 → +CPU_D4{MUL23}
- TF5 → +CPU_D4{TF5}

MMC5A Only

CPU_D5_IN_PIN → E7 → /CPU_D5{E7}
- G5 → G6 → +CPU_D5{G6}
  - E57 → /CPU_D5{E57}
  - E62 → +CPU_D5{E62}
  - L127 → /CPU_D5{L127}
- X23 → /CPU_D5{X23}
  - Y19 → +CPU_D5{Y19}
- AE23 → /CPU_D5{AE23}
- MUL24 → +CPU_D5{MUL24}
- TF6 → +CPU_D5{TF6}

MMC5A Only

CPU_D6_IN_PIN → E6 → /CPU_D6{E6}
- G3 → G4 → +CPU_D6{G4}
  - E58 → /CPU_D6{E58}
  - E63 → +CPU_D6{E63}
  - L128 → /CPU_D6{L128}
- X24 → /CPU_D6{X24}
  - P7 → +CPU_D6{P7}
  - Y25 → +CPU_D6{Y25}
- AE22 → /CPU_D6{AE22}
- MUL25 → +CPU_D6{MUL25}
- TF7 → +CPU_D6{TF7}

MMC5A Only

CPU_D7_IN_PIN → E8 → /CPU_D7{E8}
- G1 → G2 → +CPU_D7{G2}
  - E59 → /CPU_D7{E59}
  - E64 → +CPU_D7{E64}
  - L129 → /CPU_D7{L129}
- X25 → /CPU_D7{X25}
  - P5 → +CPU_D7{P5}
  - Y26 → +CPU_D7{Y26}
- AE21 → /CPU_D7{AE21}
- MUL26 → +CPU_D7{MUL26}
- TF8 → +CPU_D7{TF8}

MMC5A Only

Multiplexing Data to be Driven onto the CPU Data Bus

+CPU_ExRAM_Access{BL51} → BL61.1 → BL61.2
+ExRAM_Mode_$0x{BL48} → BL61.2

BL61.2 → BK53 (NAND_2)
- ExRAM_D0_Read{BJ12(a)}
- $5206.0_Read_Multiply_Result_D8{BL78}
- $5204.0_Read_Invalid_A0_M2_Sequence{BL52}
- $5205.0_Read_Multiply_Result_D0{BL70}
- $5015.0_Read_Pulse_1_Status_Bit{BL21}

BK53 → BK54 (NAND_2) → CPU_D0_OUT_PIN

BK55 (AOI_24)
- $5015.0_Read_Pulse_1_Status_Bit{BL21}
- $5205.0_Read_Multiply_Result_D0{BL70}
- $5204.0_Read_Invalid_A0_M2_Sequence{BL52}
- $5206.0_Read_Multiply_Result_D8{BL78}

BK58 (NAND_2)
- ExRAM_D1_Read{BJ12(b)}
- $5206.1_Read_Multiply_Result_D9{BL77}
- $5205.1_Read_Multiply_Result_D1{BL69}
- $5015.1_Read_Pulse_2_Status_Bit{BL20}

BK58 → BK57 (NAND_2) → CPU_D1_OUT_PIN

BK56 (AOI_23)
- $5015.1_Read_Pulse_2_Status_Bit{BL20}
- $5205.1_Read_Multiply_Result_D1{BL69}
- $5206.1_Read_Multiply_Result_D9{BL77}

BK59 (NAND_2)
- ExRAM_D2_Read{BJ12(c)}
- $5206.2_Read_Multiply_Result_D10{BL76}
- $5205.2_Read_Multiply_Result_D2{BL68}

BK59 → BK60 (NAND_2) → CPU_D2_OUT_PIN

BK61 (AOI_2)
- $5205.2_Read_Multiply_Result_D2{BL68}
- $5206.2_Read_Multiply_Result_D10{BL76}

BK64 (NAND_2)
- ExRAM_D3_Read{BJ12(d)}
- $5206.3_Read_Multiply_Result_D11{BL75}
- $5205.3_Read_Multiply_Result_D3{BL67}

BK64 → BK63 (NAND_2) → CPU_D3_OUT_PIN

BK62 (AOI_2)
- $5205.3_Read_Multiply_Result_D3{BL67}
- $5206.3_Read_Multiply_Result_D11{BL75}

BK65 (NAND_2)
- ExRAM_D4_Read{BJ12(e)}
- $5206.4_Read_Multiply_Result_D12{BL74}
- $5205.4_Read_Multiply_Result_D4{BL66}

BK65 → BK66 (NAND_2) → CPU_D4_OUT_PIN

BK67 (AOI_2)
- $5205.4_Read_Multiply_Result_D4{BL66}
- $5206.4_Read_Multiply_Result_D12{BL74}

BK70 (NAND_2)
- ExRAM_D5_Read{BJ12(f)}
- $5206.5_Read_Multiply_Result_D13{BL73}
- $5205.5_Read_Multiply_Result_D5{BL65}

BK70 → BK69 (NAND_2) → CPU_D5_OUT_PIN

BK68 (AOI_2)
- $5205.5_Read_Multiply_Result_D5{BL65}
- $5206.5_Read_Multiply_Result_D13{BL73}

BK71 (NAND_2)
- ExRAM_D6_Read{BJ12(g)}
- $5206.6_Read_Multiply_Result_D14{BL64}
- $5205.6_Read_Multiply_Result_D6{BL72}
- $5010.6_Read_In_Frame{BL80}

BK71 → BK72 (NAND_2) → CPU_D6_OUT_PIN

BK73 (AOI_23)
- $5010.6_Read_In_Frame{BL80}
- $5205.6_Read_Multiply_Result_D6{BL72}
- $5206.6_Read_Multiply_Result_D14{BL64}

BK76 (NAND_2)
- ExRAM_D7_Read{BJ12(h)}
- $5206.7_Read_Multiply_Result_D15{BL63}
- $5205.7_Read_Multiply_Result_D7{BL71}
- $5204.7_Read_Scanline_IRQ_Pending{BL79}
- $5010.7_Read_PCM_IRQ_Pending{BL62}

BK76 → BK75 (NAND_2) → CPU_D7_OUT_PIN

BK74 (AOI_24)
- $5010.7_Read_PCM_IRQ_Pending{BL62}
- $5204.7_Read_Scanline_IRQ_Pending{BL79}
- $5205.7_Read_Multiply_Result_D7{BL71}
- $5206.7_Read_Multiply_Result_D15{BL63}

$5015.0_Read_Pulse_1_Status_Bit{B106} → BL21 → $5015.0_Read_Pulse_1_Status_Bit{BL21}
$5015.1_Read_Pulse_2_Status_Bit{B104} → BL20 → $5015.1_Read_Pulse_2_Status_Bit{BL20}
+Invalid_A0_M2_Sequence{BM25} → BL52 → $5204.0_Read_Invalid_A0_M2_Sequence{BL52}
+PCM_IRQ_Status_Bit{K114} → BL62 → $5010.7_Read_PCM_IRQ_Pending{BL62}
/Scanline_IRQ_Status{X28} → BL79 → $5204.7_Read_Scanline_IRQ_Pending{BL79}
+In_Frame{Z21} → BL80 → $5010.6_Read_In_Frame{BL80}

Decides When to Drive the CPU Data Bus

+CPU_$5015_Access{L118} → BL22 → +CPU_$5015_Access{BL22} → BP1
+CPU_$5204_Access{L88} → BL56 → +CPU_$5204_Access{BL56} → BK48 → BP1
+CPU_$5205_Access{L91} → BL55 → +CPU_$5205_Access{BL55} → BP3
+CPU_$5206_Access{L92} → BL54 → +CPU_$5206_Access{BL54} → BP4

BP1, BP3, BP4 → BP2 (NAND_4) → /$5105,5204,5205,5206_Access{BP2}

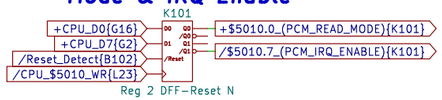
+CPU_ExRAM_Access{L117} → BL51 → +CPU_ExRAM_Access{BL51} → BK52
+ExRAM_Mode_$0x{BL96} → BL48 → +ExRAM_Mode_$0x{BL48} → BK50

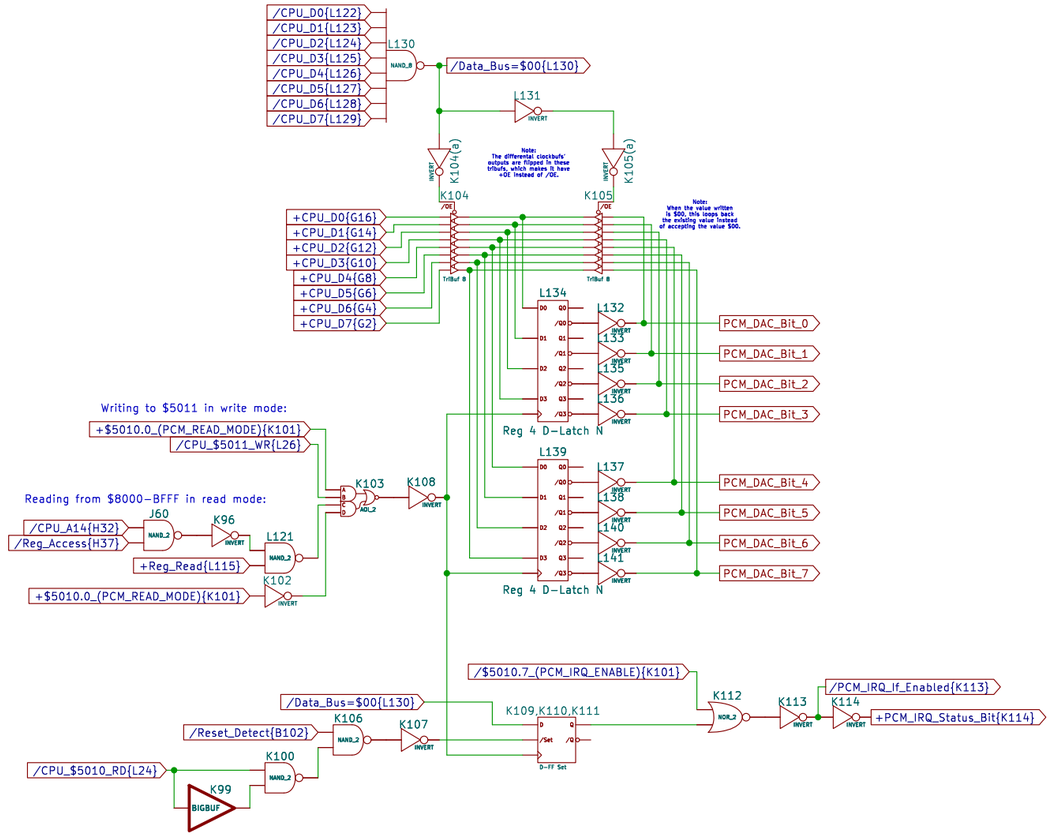
BK52, BK50 → BK51 (OAI_2) → BK46 (OAI_2) → BK45 (INVERT) → CPU_Dx_Direction{BK45}

+CPU_$5010_Access{L22} → BL53 → +CPU_$5010_Access{BL53} → BK49
/Reg_Read{K2} → BL50 → BK47

Sheet: /CPU Data Bus/	
File: cpu_data_bus.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 8/27

$5010 Register
Mode & IRQ Enable





Sheet: /PCM DAC/	
File: pcm_dac.kicad_sch	
Title:	
Size: USLedger | Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 9/27

Scanline Start Detection

+Use Internal Scanline Detection

+Use External Scanline Detection

+Test_Mode_PIN_92{Pin1nv} → AA1 (INVERT,3-State Buffer)

/Test_Input_PIN_29{Pin1nv} → AA4 (INVERT)

External Scanline Start

High when the present PPU address is from $2xxx and matches the previous read.

AA5 (NOR_4) ← +PPU_A13_B{AB21}, +PPU_A12_B{AB20}, +PPU_A11_B{AB15}, +PPU_A10_B{AB14}, +PPU_A9{AB9}, +PPU_A8{AB8}, +PPU_A7{AB5}, +PPU_A6{AC10}, +PPU_A5{AC11}, +PPU_A4{AC16}, +PPU_A3{AC17}, +PPU_A2{AC22}, +PPU_A1{AC23}, +PPU_A0{AC28}

+PPU_A13_B{AB21} → AB22 (D-FF Reset) → AA21 (NAND_2) → AA20 (INVERT)

+PPU_A12_B{AB20} → AB19 (D-FF Reset) → AA18 (NOR_2)

+PPU_A11_B{AB15} → AB16 (D-FF Reset) → AA17 (XNOR_2)

+PPU_A10_B{AB14} → AB13 (D-FF Reset) → AA14 (XNOR_2)

AA16 (NAND_4) ← AA17, AA14

+PPU_A9{AB9} → AB10 (D-FF Reset) → AA13 (XNOR_2)

+PPU_A8{AB8} → AB7 (D-FF Reset) → AA12 (XNOR_2)

+PPU_A7{AB5} → AB6 (D-FF Reset) → AA9 (XNOR_2)

AA10 (NAND_4) ← AA13, AA12, AA9, AA8

/PPU_RD_B{AB4} → AA7 (INVERT)

/Exit_In_Frame{AC6} → AA6 (INVERT)

/PPU_RD_B{AC7} → AB3 (INVERT)

+PPU_A6{AC10} → AC9 (D-FF Reset) → AA8 (XNOR_2)

+PPU_A5{AC11} → AC12 (D-FF Reset) → AA11 (XNOR_2)

+PPU_A4{AC16} → AC15 (D-FF Reset) → AB11 (XNOR_2)

+PPU_A3{AC17} → AC18 (D-FF Reset) → AB12 (XNOR_2)

+PPU_A2{AC22} → AC21 (D-FF Reset) → AB17 (XNOR_2)

+PPU_A1{AC23} → AC24 (D-FF Reset) → AB18 (XNOR_2)

+PPU_A0{AC28} → AC27 (D-FF Reset) → AA22 (XNOR_2)

AA15 (NAND_4) ← AA11, AB11, AB12, AB17, AB18

AA19 (NAND_2) ← AA15, AA16

When not in frame, the previous PPU read address always looks like $0000. Since A13, A12 are looking at range $2xxx instead of matches, $0xxx won't match $2xxx. Therefore, scanlines don't get detected when using PPUDATA to read from $0000 during v-blank.

AB2 (SMR Reg 3) → AB2(a) (INVERT) → AB1 (NAND_4) ← +PPU_RD_B{W1}

Low when the 3 most recent PPU reads were from $2xxx and all matched.

AA2 (AOI_2) ← AA1, AB1, AA4

AA3 (INVERT) ← AA2 → /Scanline_Start_Detected{AA3}

R2 (BUFFER) ← AA3 → /Scanline_Start_Detected{R2}

AC2 (Counter 2 Reset) ← M2_CLK_BUF1, /PPU_RD_B{AC1}

AC6 (NAND_2) ← AC2 → /Exit_In_Frame{AC6}

Scanline End Detection

Increment_Scanline_Fetch_Count{S6}, +PPU_Cycle_336{R4} → R6 (NAND_2) → R7 (INVERT) → R8 (INVERT) → Scanline_End_Detected_Internal{R8}

Scanline_End_Detected_Internal{R8}, /Test_Input_PIN_30{Pin1nv} → R1 (AOI_3) → P3 (INVERT) → Scanline_End_Detected{P3}

P3 → T5 (BUFFER) → Scanline_End_Detected{T5}

/Test_Input_PIN_30{Pin1nv} → P4 (INVERT)

External Scanline End

$5204 Register Scanline IRQ Enable

+CPU_D7{Y26}, /Reset_Detect{X15}, /CPU_$5204_WR{L87} → X12 (D-FF Reset) → X13 (NAND_2) → X14 (BUFFER) → +Scanline_IRQ_Pin{X14}

This signal likely needs to be active low.

$5203 Register IRQ Scanline Compare Value

Y2,7,9,14,16,21,23,28

+CPU_D0{Y4}, +CPU_D1{Y5}, +CPU_D2{Y11}, +CPU_D3{Y12}, +CPU_D4{Y18}, +CPU_D5{Y19}, +CPU_D6{Y25}, +CPU_D7{Y26} → Y1 (BUFFER)

/CPU_$5203_WR{L83} → Y1

Reg 8 D-Latch

Y3, Y6, Y10, Y13, Y17, Y20, Y24, Y27 (XNOR_2) → Y8 (NAND_4), Y22 (NAND_4) → Y15 (NOR_2)

Hi when $5203 matches the present V-Split scanline.

D'Morgan tells us that we can pretend Y8,Y15,Y22 are all AND gates.

IRQ Scanline Counter

Z2-Z18

/Scanline_Start_Detected{AA3}, +In_Frame{Z21} → Z1 (NAND_2)

/In_Frame{Z23} → Z19 (BUFFER)

Z1, Z19 → Counter 8 Reset

Z22 (NAND_4) ← Z2-Z18 → /Scanline_240_and_up{Z22}

$2001 Register Write Snooping

+CPU_D3{Y12}, +CPU_D4{Y18}, /CPU_$2001_WR{L107} → X8 (Reg 2 DFF-No-Reset N) → X9 (NAND_2)

Hi = PPU Rendering Enabled

/CPU_$5204_RD{L86} → M3 (BIGBUF) → N38 (NAND_2) → N39 (INVERT) → M2 (NAND_2)

+New_Scanline_Detected{AA3 via Z23}, /In_Frame{Z23} → Z20 (NAND_2)

M2, Z20 → X17 (NAND_2) → X26 (INVERT) → Y31 (D-FF Reset)

/Scanline_Start_Detected{AA3} → Y31

+IRQ → Y29 (NOR_2) → Y30 (INVERT) → /CPU_$5204_RD{L86}

Y31 → Y32 (INVERT) → X27 (D-FF No Reset) → X28 (INVERT) → /Scanline_IRQ_Status{X28}

/IRQ

This signal likely needs to be active hi.

$2002 Register Read Snooping

/CPU_$2002_RD{L110} → X16 (INVERT) → X4 (NAND_2)

+CPU_D7{Y26}, M2_CLK_BUF1 → X5 (D-FF No Reset)

X4, X5 → X6 (NAND_4) → X7 (INVERT) → Z23 (D-FF Reset)

/CPU_$FFFA/B_RD{L21}, /CPU_$4014_WR{L116}, /Exit_In_Frame{AC6} → X6

/Scanline_Start_Detected{AA3} → Z23

Z23 → /In_Frame{Z23}, Z21 (INVERT) → +In_Frame{Z21}

Latches $2002.7 value read by CPU for 1 CPU cycle.

$2001 Register Write Snooping

Z24 (NAND_2) ← /In_Frame{Z23}, +In_Frame{Z21}

Z24 → Z25 (INVERT) → +Rendering_In_Frame{Z25}

V39 (INVERT) → /Rendering_In_Frame{V39}

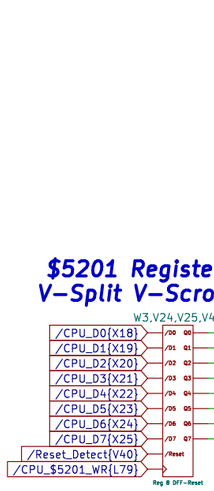
R24 (INVERT) → /Rendering_In_Frame{R24}

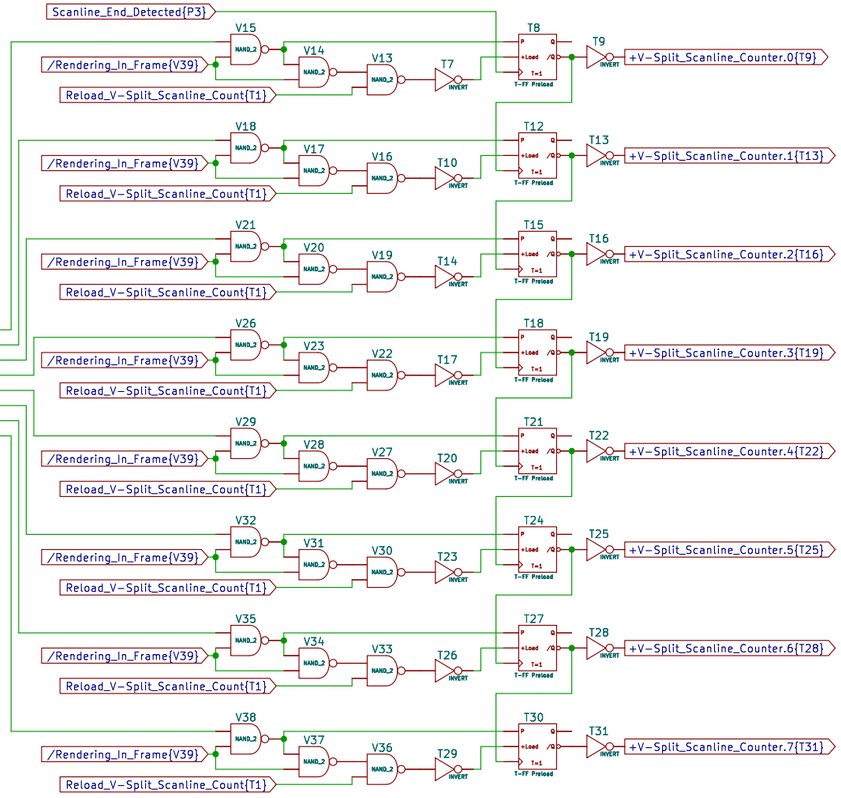
N16 (INVERT) → N9 (INVERT) → +Rendering_In_Frame{N9}

Sheet: /Scanline Detection/	
File: scanline_detection.kicad_sch	
Title:	
Size: B	Date:
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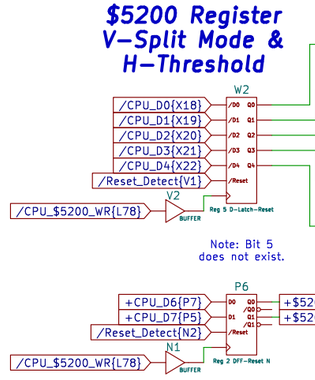
V−Split Scanline Counter

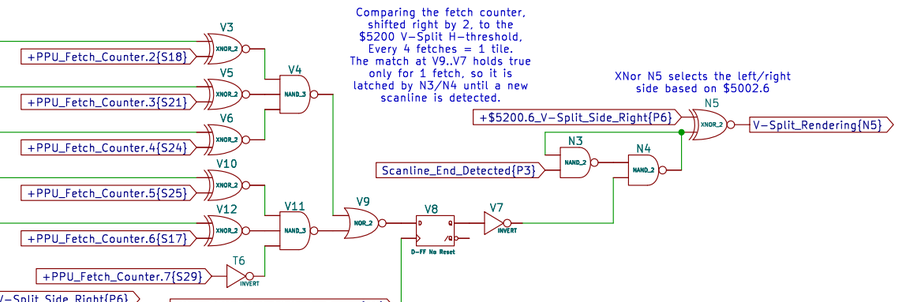
$5201 Register V−Split V−Scroll

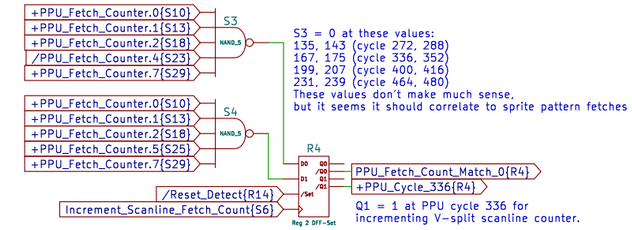


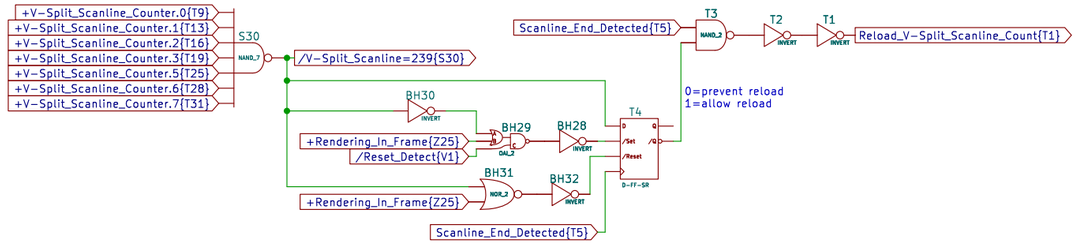


$5200 Register V−Split Mode & H−Threshold



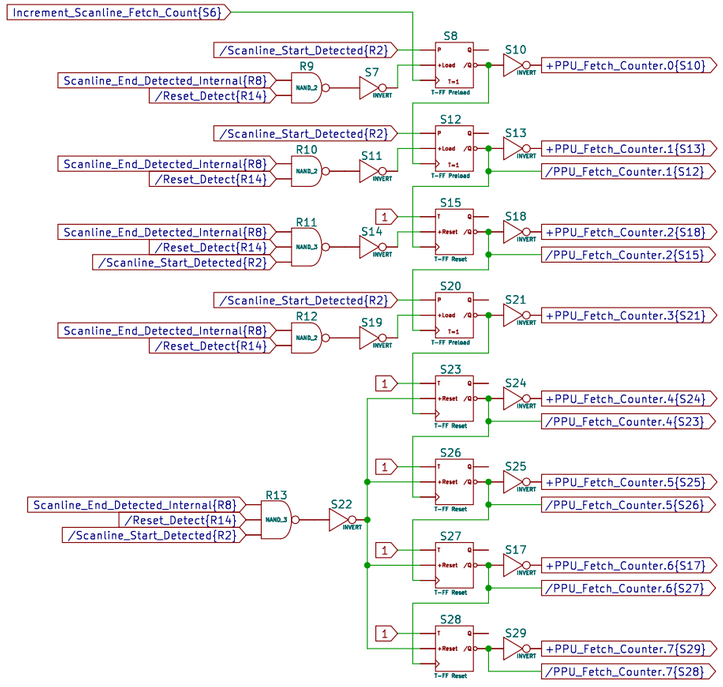




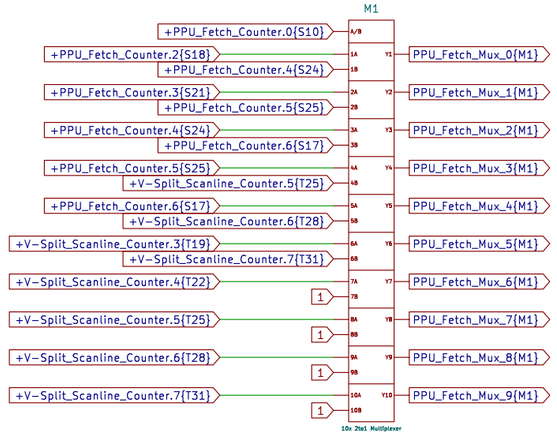


Sheet: /V−Split/		
File: v_split.kicad_sch		
Title:		
Size: B	Date:	**Rev:**
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PPU Fetch Counter

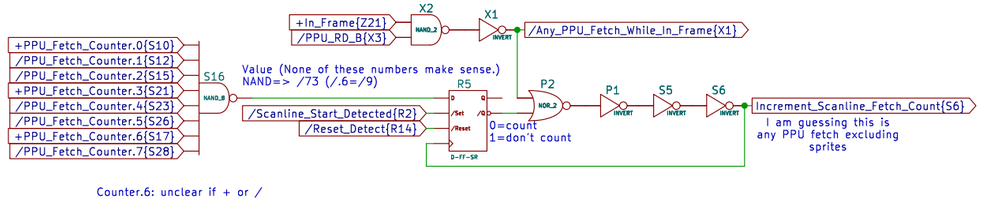


ExRAM Attribute Fetch Mux



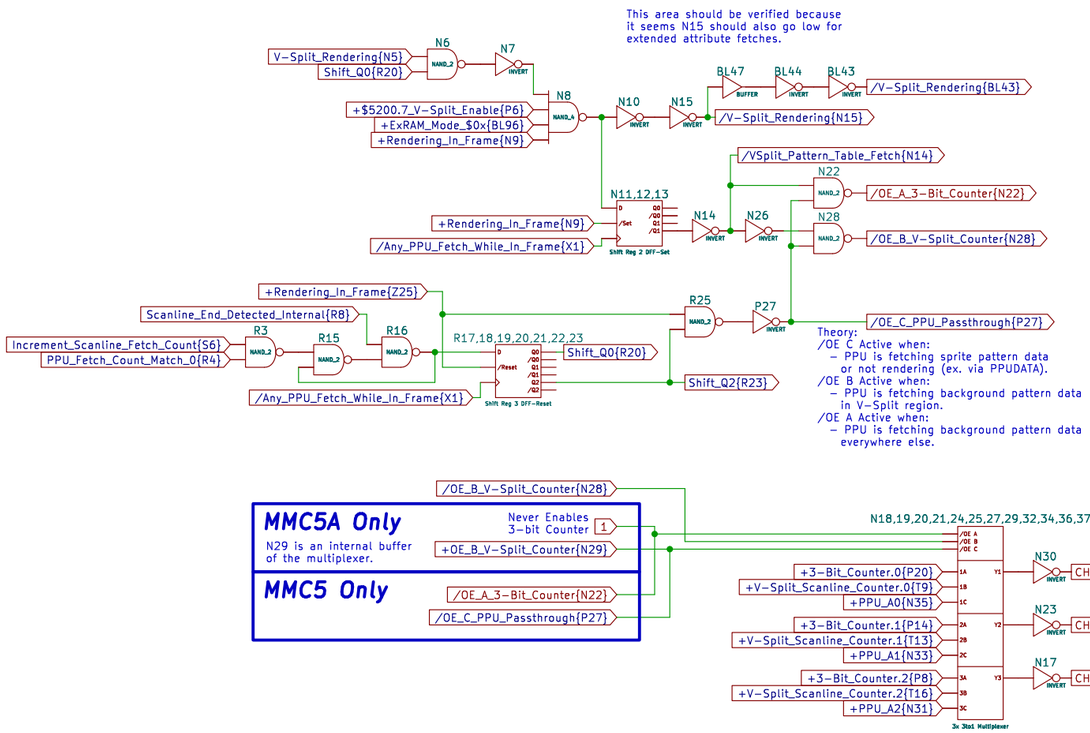
The output of this mux is the ExRAM address for the data to be substituted for the PPU.

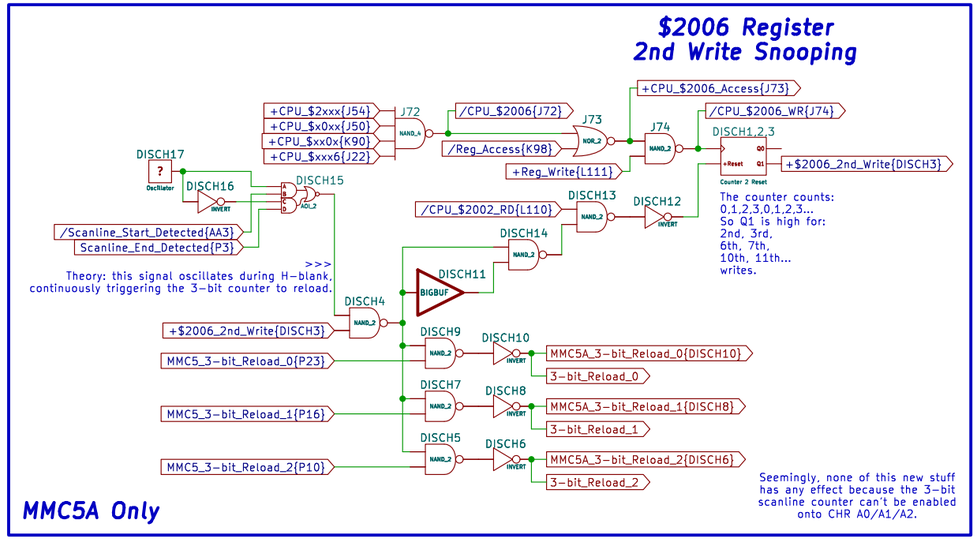
See sheet 14.

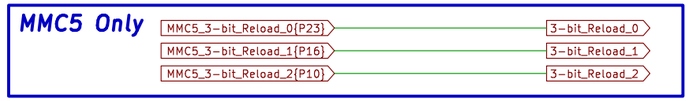


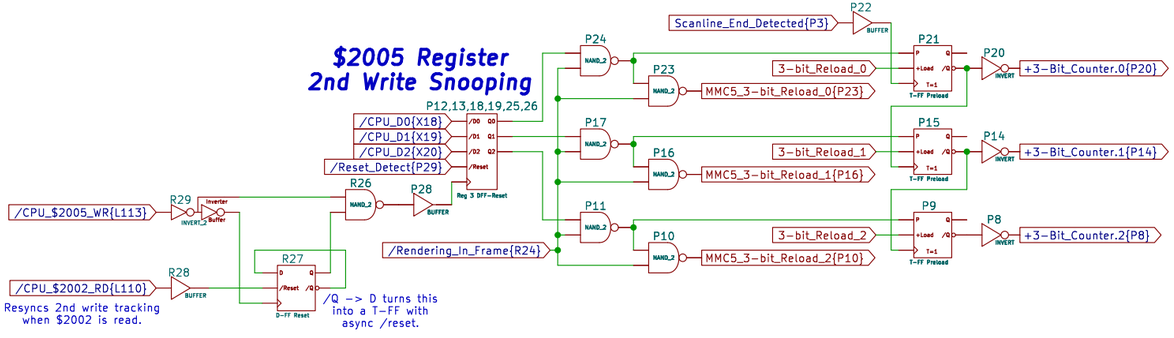
Counter.6: unclear if + or /

Sheet: /PPU Fetch Counting/	
File: ppu_fetch_counting.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 12/27









Sheet: /CHR A0,A1,A2/	
File: chr_a0_a1_a2.kicad_sch	
Title:	
Size: USLedger Date:	**Rev:**
KiCad E.D.A. 9.0.7	Id: 13/27

$5104 Register ExRAM Mode

CPU_D0_IN_PIN
CPU_D1_IN_PIN
/Reset_Detect{BL97}
/CPU_$5104_WR{L37}
BK82
Reg 2 DFF-Set
BK79
BK87
+ExRAM_Mode_%10{BK87}
BK80
BL49
/ExRAM_Mode_%01{BK80}
/ExRAM_Mode_%01{BL49}
BK81
BL96
/ExRAM_Mode_%00{BK81}
Appears to go nowhere.
+ExRAM_Mode_$0x{BL96}

Tracks whether $5104 has been written since reset detection

/Reset_Detect{RAX12}
M2_CLK_BUF1
+Invalid_A0_M2_Sequence{BN1}
Syncs SL3 to M2
BL89
BL90
BL91
BL98
D-FF Reset
SR Latch
BL94
BL92
BL93
BN2
BIGBUF

There appears to be a selectable delay jumper, connected at BM6.

BM10 BM9 BM8 BM7 BM6 BM5 BM4 BM3 BM2 BM1
BUFFER

+ExRAM_Mode_$0x{BL96}
BL86
Inverter Buffer
+PPU_RD_A_PIN{PinInv}
+PPU_WR_PIN{PinInv}
BL82
BL88
BL85
BL84
BL83
PPU_RW_Mode_%0x_Access{BL83}
Not sure about polarity; this may be Mode %1x Access, and active low or high.
BL87
BL81
BIGBUF

PPU is fetching from nametable or attribute ($2000+) and V-Split is not rendering and ExRAM mode is %0x (a PPU mode)

+PPU_A13_A{BK97}
+ExRAM_Mode_$0x{BL96}
BK94
BK95
+Drive_ExRAM_Ax_From_PPU_Ax{BK95}

/V-Split_Rendering{N15}
BK92
BK93
BK91

PPU is fetching from nametable or attribute ($2000+) and V-Split is rendering and ExRAM mode is %0x (a PPU mode)

+Drive_ExRAM_Ax_For_Substituted_Fetch{BK91}

PPU is fetching from pattern table (below $2000) or ExRAM mode is %1x (a CPU mode)

BK96
BJ51
+Drive_ExRAM_Ax_From_CPU_Ax{BJ51}

Note:
Maybe N15 means /Extended Attribute Fetch OR V-Split Rendering?
N15 seems to directly depend on V-Split enable register bit though.

PPU_Fetch_Mux_0{M1} PPU_A0_Pin BJ50 ExRAM_A0{BJ50}
PPU_Fetch_Mux_1{M1} PPU_A1_Pin BJ49 ExRAM_A1{BJ49}
PPU_Fetch_Mux_2{M1} PPU_A2_Pin BJ48 ExRAM_A2{BJ48}
PPU_Fetch_Mux_3{M1} PPU_A3_Pin BJ47 ExRAM_A3{BJ47}
PPU_Fetch_Mux_4{M1} PPU_A4_Pin BJ46 ExRAM_A4{BJ46}
PPU_Fetch_Mux_5{M1} PPU_A5_Pin BJ45 ExRAM_A5{BJ45}
PPU_Fetch_Mux_6{M1} PPU_A6_Pin BJ44 ExRAM_A6{BJ44}
PPU_Fetch_Mux_7{M1} PPU_A7_Pin BJ43 ExRAM_A7{BJ43}
PPU_Fetch_Mux_8{M1} PPU_A8_Pin BJ42 ExRAM_A8{BJ42}
PPU_Fetch_Mux_9{M1} PPU_A9_Pin BJ41 ExRAM_A9{BJ41}
NAND Mux 3

CPU_A0_Pin
CPU_A1_Pin
CPU_A2_Pin
CPU_A3_Pin
CPU_A4_Pin
CPU_A5_Pin
CPU_A6_Pin
CPU_A7_Pin
CPU_A8_Pin
CPU_A9_Pin
BJ40
Reg 10 D-Latch

+CPU_ExRAM_Access{L117}
+Reg_Write{BK88}
BK85
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BK86
Writing to ExRAM via $5C00-5FFF in mode %10
+ExRAM_Mode_%10{BK87}

Writing to an ExRAM nametable via PPU via $2000

+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL96}
+PPU_WR_PIN{PinInv}
BK83
BK84
+ExRAM_WR{BK84}
This is the write signal for the ExRAM itself.

BN4
BIGBUF
+ExRAM_Mode_$0x{BL96}
BL105
BL106
BL109
/Q{BL109}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
BL108
BL99
/Q{BL99}
+PPU_A13_A{BK97}
BL102
/Reset_Detect{RAX12}
BL101
BL100
BN3
BIGBUF
BL107
+PPU_RD_A_PIN{PinInv}

/Q{BL109}
/CPU_Write_To_ExRAM_Any_Mode{BK85}
+ExRAM_Mode_$0x{BL96}
BK89

MMC5 Only
+PPU_RD_A_PIN{PinInv}
MMC5A Only
/Q{BL99}
BK90
+ExRAM_Mode_$0x{BL96}

CPU_D0_IN_PIN
CPU_D1_IN_PIN
CPU_D2_IN_PIN
CPU_D3_IN_PIN
CPU_D4_IN_PIN
CPU_D5_IN_PIN
CPU_D6_IN_PIN
CPU_D7_IN_PIN
BK78
Reg 8 D-Latch
BK77
Reg 8 D-Latch
CPU_D0_In_Latched{BK77}
CPU_D1_In_Latched{BK77}
CPU_D2_In_Latched{BK77}
CPU_D3_In_Latched{BK77}
CPU_D4_In_Latched{BK77}
CPU_D5_In_Latched{BK77}
CPU_D6_In_Latched{BK77}
CPU_D7_In_Latched{BK77}

ExRAM_D0_Read BJ12(a) ExRAM_D0_Read{BJ12(a)} /ExRAM_Read_D0_Latch{BJ11}
ExRAM_D1_Read BJ12(b) ExRAM_D1_Read{BJ12(b)} /ExRAM_Read_D1_Latch{BJ11}
ExRAM_D2_Read BJ12(c) ExRAM_D2_Read{BJ12(c)} /ExRAM_Read_D2_Latch{BJ11}
ExRAM_D3_Read BJ12(d) ExRAM_D3_Read{BJ12(d)} /ExRAM_Read_D3_Latch{BJ11}
ExRAM_D4_Read BJ12(e) ExRAM_D4_Read{BJ12(e)} /ExRAM_Read_D4_Latch{BJ11}
ExRAM_D5_Read BJ12(f) ExRAM_D5_Read{BJ12(f)} /ExRAM_Read_D5_Latch{BJ11}
ExRAM_D6_Read BJ12(g) ExRAM_D6_Read{BJ12(g)} +ExRAM_Read_D6_Latch{BJ11}
ExRAM_D7_Read BJ12(h) ExRAM_D7_Read{BJ12(h)} +ExRAM_Read_D7_Latch{BJ11}
BUFFER
BJ11
Reg 8 D-Latch N

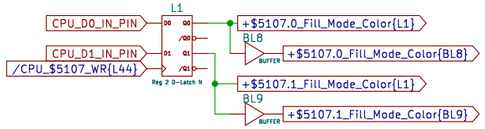
+PPU_RD_A{BJ8}
+PPU_Fetch_Count.1{S13}
BJ9
BJ10
BJ7
+PPU_Fetch_Count.0{S10}

/Scanline_Counter_Reset{Z25}
+ExRAM_Mode_$0x{BL96}
BL95
BL103
BL104
BJ39b
Inverter Buffer

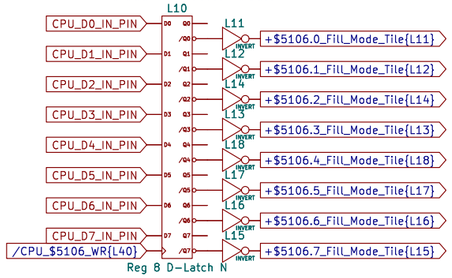
CPU_D0_In_Latched{BK77} PPU_D0_In{BigBuf} BJ39 ExRAM_D0_Write{BJ39}
CPU_D1_In_Latched{BK77} PPU_D1_In{BigBuf} BJ38 ExRAM_D1_Write{BJ38}
CPU_D2_In_Latched{BK77} PPU_D2_In{BigBuf} BJ37 ExRAM_D2_Write{BJ37}
CPU_D3_In_Latched{BK77} PPU_D3_In{BigBuf} BJ36 ExRAM_D3_Write{BJ36}
CPU_D4_In_Latched{BK77} PPU_D4_In{BigBuf} BJ35 ExRAM_D4_Write{BJ35}
CPU_D5_In_Latched{BK77} PPU_D5_In{BigBuf} BJ34 ExRAM_D5_Write{BJ34}
CPU_D6_In_Latched{BK77} PPU_D6_In{BigBuf} BJ33 ExRAM_D6_Write{BJ33}
CPU_D7_In_Latched{BK77} PPU_D7_In{BigBuf} BJ32 ExRAM_D7_Write{BJ32}
NAND Mux 2

Sheet: /ExRAM Mode/
File: exram_mode.kicad_sch
Title:
Size: USLedger | Date:
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Rev:
Id: 14/27

$5107 Register
Fill-Mode Color



$5106 Register
Fill-Mode Tile



Sheet: /Fill Mode/
File: fill_mode.kicad_sch

Title:

Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 15/27

5-Bit to 7-Bit Pulse Length Lookup Table

See spreadsheet for decoded values:
"pulse length lookup table.xlsx"

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
D46 NAND_5
D45 INVERT
+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
D48 NAND_4
D47 INVERT
C16 NOR_2
C15 INVERT
Pulse_LUT_D6{C15}

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
D50 NAND_5
D49 INVERT
+CPU_D7{E64}
+CPU_D6{E63}
/CPU_D5{E57}
/CPU_D3{E55}
D52 NAND_4
D51 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
D54 NAND_5
D53 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E56}
+CPU_D3{E60}
D56 NAND_5
D55 INVERT
C18 NOR_4
C17 INVERT
Pulse_LUT_D5{C17}

+CPU_D7{E64}
+CPU_D6{E63}
+CPU_D5{E62}
+CPU_D4{E61}
/CPU_D3{E55}
D57 NAND_5
D58 INVERT
+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D3{E55}
D59 NAND_4
D60 INVERT
/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
D61 NAND_4
D62 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
/CPU_D3{E55}
D63 NAND_5
D64 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
D65 NAND_5
D66 INVERT
C20 NOR_5
C19 INVERT
Pulse_LUT_D4{C19}

+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
D67 NAND_5
D68 INVERT
/CPU_D7{E59}
+CPU_D4{E61}
/CPU_D3{E55}
D69 NAND_3
D70 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
+CPU_D3{E60}
D71 NAND_5
D72 INVERT
+CPU_D7{E64}
+CPU_D3{E60}
D73 NAND_2
D74 INVERT
C22 NOR_5
C21 INVERT
Pulse_LUT_D3{C21}

+CPU_D7{E64}
/CPU_D6{E58}
+CPU_D5{E62}
/CPU_D4{E56}
D75 NAND_4
D76 INVERT

+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
D77 NAND_3
D78 INVERT
/CPU_D7{E59}
+CPU_D5{E62}
/CPU_D4{E56}
/CPU_D3{E55}
D79 NAND_4
D80 INVERT
+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D3{E55}
D81 NAND_4
D82 INVERT
/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
D83 NAND_3
D84 INVERT
+CPU_D6{E63}
+CPU_D3{E60}
D85 NAND_2
D86 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
D87 NAND_4
D88 INVERT
C24 NOR_6
C23 INVERT
Pulse_LUT_D2{C23}

+CPU_D7{E64}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E56}
/CPU_D3{E55}
D89 NAND_5
D90 INVERT
/CPU_D7{E59}
+CPU_D6{E63}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
D91 NAND_5
D92 INVERT
/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D4{E56}
/CPU_D3{E55}
D93 NAND_5
D94 INVERT
+CPU_D5{E62}
+CPU_D3{E60}
D95 NAND_3
D96 INVERT
C26 NOR_6
C25 INVERT
Pulse_LUT_D1{C25}

/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E61}
/CPU_D3{E55}
D97 NAND_5
D98 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
+CPU_D4{E56}
+CPU_D3{E60}
D99 NAND_5
D100 INVERT
/CPU_D7{E59}
+CPU_D6{E63}
+CPU_D5{E62}
/CPU_D3{E55}
D101 NAND_4
D102 INVERT
/CPU_D7{E59}
/CPU_D6{E58}
/CPU_D5{E57}
/CPU_D4{E56}
D103 NAND_4
D104 INVERT
+CPU_D4{E61}
+CPU_D3{E60}
D105 NAND_2
D106 INVERT
C28 NOR_3
C27 INVERT
Pulse_LUT_D0{C27}

$5003 Register
Pulse 1 Length Counter

+$5000.5_Length_Halt{G85}
B44 INVERT
B43 NAND_2
Pulse_1_L_Restart{C31(b)}
B42 NOR_2
Pulse_Length_Clock_SL3(Reset_Aware){A75}
241.08Hz

C11
Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
P0 P1 P2 P3 Q0 Q1 Q2 Q3
Ripple Out
/Load
Counter 4 Preload

C12
Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
/CPU_$5003_WR{K16}
Ripple Out
Pulse_1_L_Rollover{C12}
/Load
Counter 4 Preload

These counters count "down"
the preloaded period value.

This means either the outputs
or preload inputs are inverted.

$5007 Register
Pulse 2 Length Counter

+$5004.5_Length_Halt{E68}
C8 INVERT
C9 NAND_2
Pulse_2_L_Restart{C31(a)}
C10 NOR_2

D38
Pulse_LUT_D0{C27}
Pulse_LUT_D1{C25}
Pulse_LUT_D2{C23}
Pulse_LUT_D3{C21}
P0 P1 P2 P3 Q0 Q1 Q2 Q3
Ripple Out
/Load
Counter 4 Preload

D39
Pulse_LUT_D4{C19}
Pulse_LUT_D5{C17}
Pulse_LUT_D6{C15}
+Reset_Detect{D40}
/CPU_$5007_WR{K80}
Ripple Out
Pulse_2_L_Rollover{D39}
/Load
Counter 4 Preload

Pulse Channel Clock Source

1.789773 MHz
+M2_CLK_BUF2{B109}
A56 NOR_2
A57 NOR_2
894.886 kHz
<25% duty, depends on M2 Duty

1
/Reset_Detect{A63}
A55 T Q +Reset /Q T-FF Reset
894.886 kHz
50% duty

The reset inputs of ALL these
counters seem to be /Reset.
Hard to see why this acts different
than T-FF counters elsewhere.

A66 Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 +Reset Counter 8 Reset
/Reset_Detect{A63}
A65 resets
Q7 of A66.
A66.2 INVERT
447.443 kHz
50% duty
3.49565 kHz
50% duty
A66.3 INVERT
A66.6 INVERT
A66.7 INVERT
A68 INVERT

A67 Q0 Q1 Q2 Q3 +Reset Counter 5 Reset
Important:
A63 resets
Q4 of A67.
/Reset_Detect{A63}
A64 NAND_2
A65 INVERT
+Reset_Detect{A62}
A66.5 NOR_3
always 0 when reset detected,
without reset, only 0 when value=3712
A66.4 NAND_5
/Value
3712

A66.2 in this logic can only
create a phase shift and
not affect the period
since it doesn't cause A66
to reset.

A60 INVERT
A61 INVERT
Pulse_Timer_Clock_Fast{A61}
894.886 kHz
A58 INVERT
A59 INVERT
Pulse_Timer_Clock_Fast{A59}

241.08 Hz
Duty seems thin
A78 ADJ_2
241.08 Hz or
Pin 98 Input
B108 INVERT
A70 NAND_2
A71 INVERT
A72 INVERT
Pulse_Timer_Clock_SL3{A72}
E51 INVERT
/Pulse_Timer_Clock_SL3{E51}
H87 INVERT
/Pulse_Timer_Clock_SL3{H87}
A73 NAND_2
A74 INVERT
A75 INVERT
Pulse_Length_Clock_SL3{Reset_Aware}{A75}

All of these outputs are
241.08 Hz
-or-
Pin 98 Input

/Q4 is high for the first
240Hz period after reset
has stopped being detected.
A83 INVERT
B107 INVERT
A69 INVERT
A79 INVERT
A81 INVERT
A80 INVERT
A77 ADJ_2
A82 INVERT
0=Reset Held By A67.Q4 or Pin 30

+Test_Mode_PIN_92{Pininv}
/Test_Input_PIN_30{Pininv}

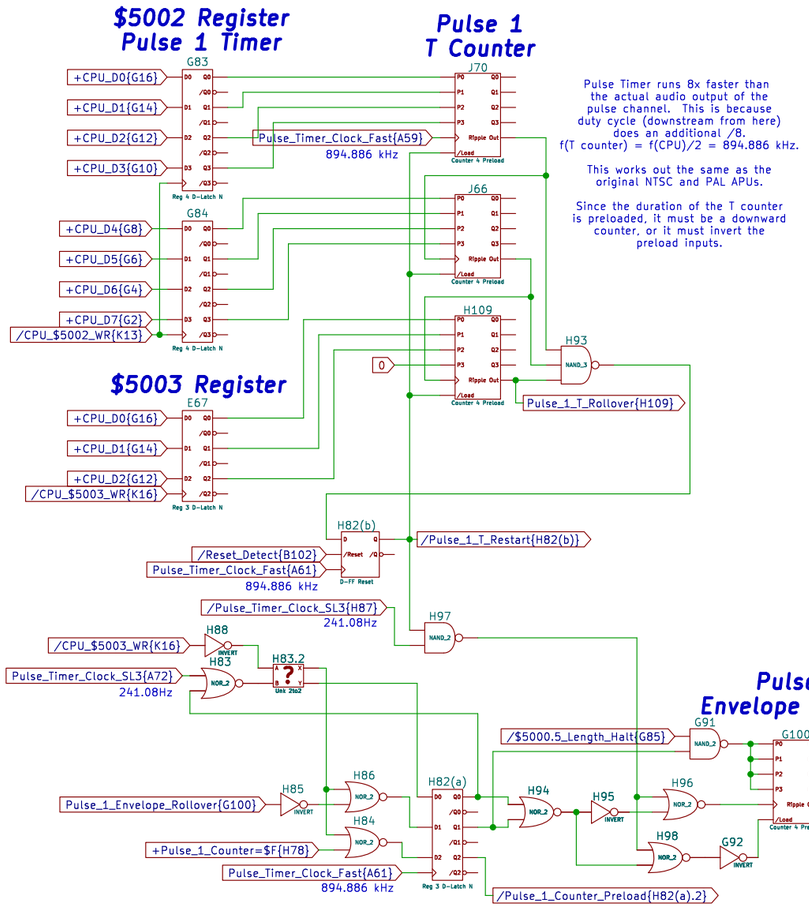
MMC5 Only
+SL3_PIN{PinBuf}

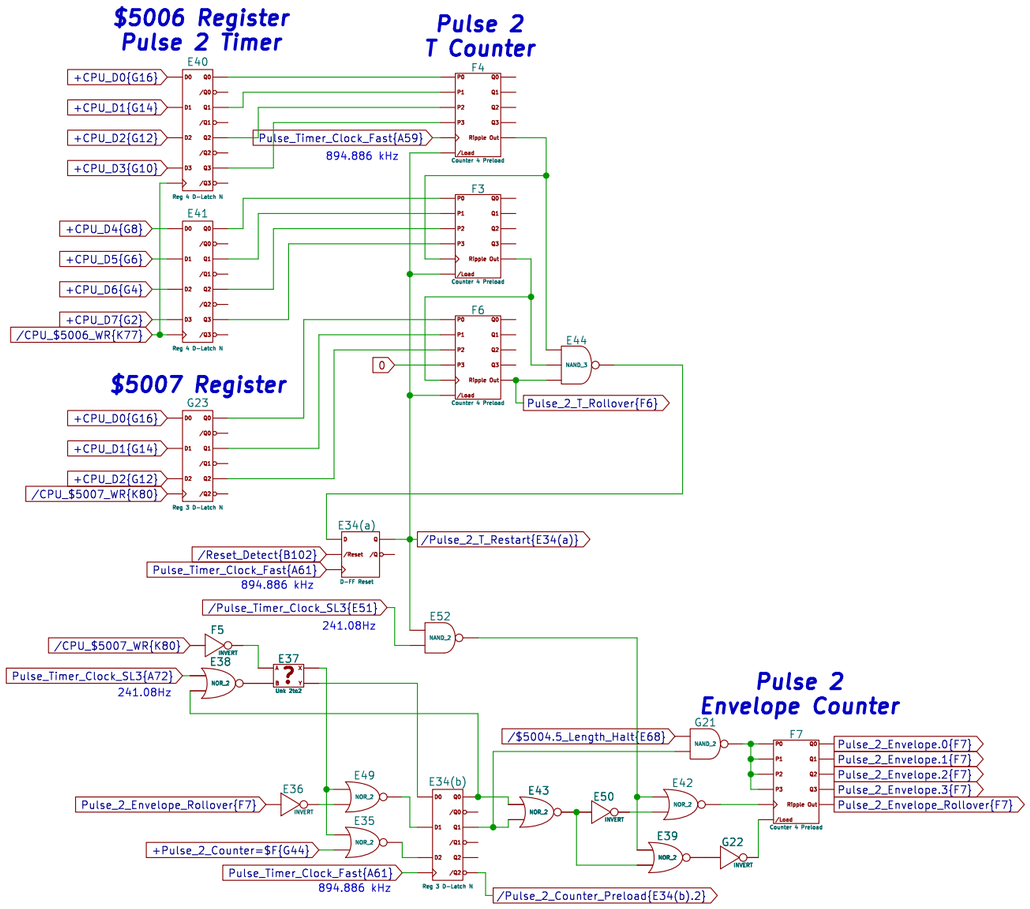
MMC5A Only
/Invalid_A0_M2_Sequence{BM11}

Sheet: /Pulse Length Lookup Table/
File: pulse_length_lookup_table.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 16/27



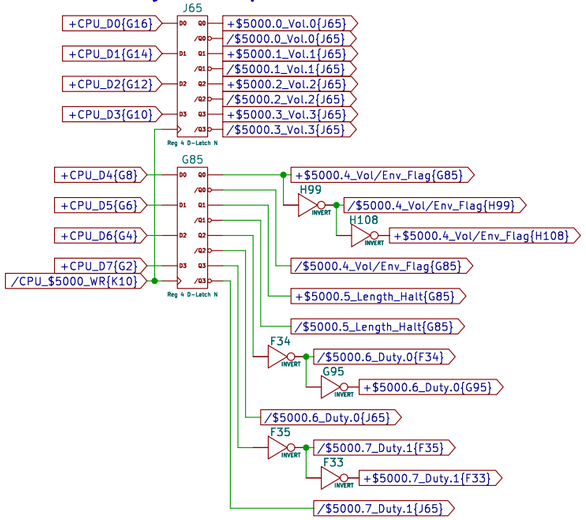


Sheet: /Pulse Timer/
File: pulse_timer.kicad_sch

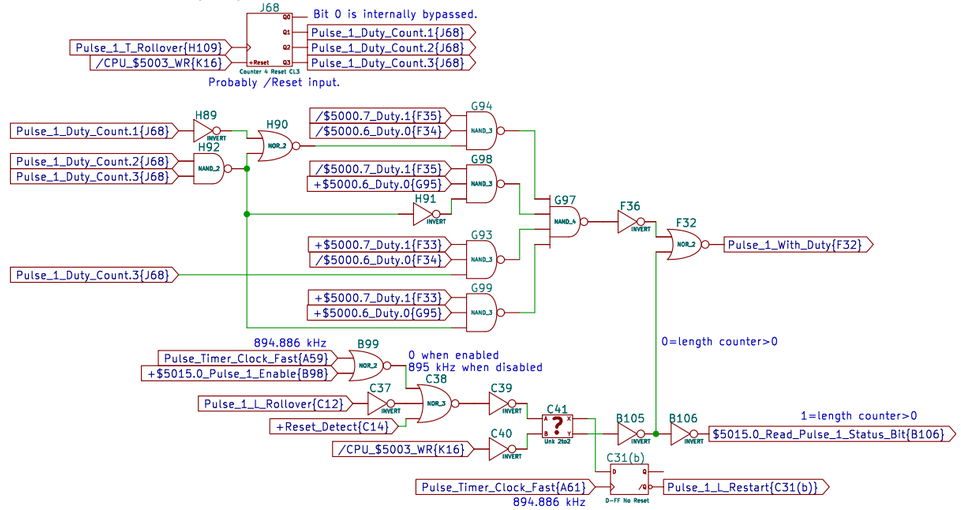
Title:

Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 17/27

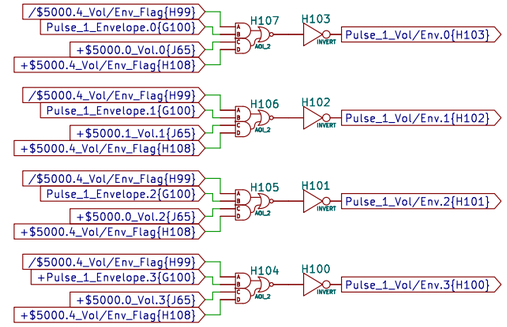
$5000 Register
Pulse 1 Duty Envelope Volume



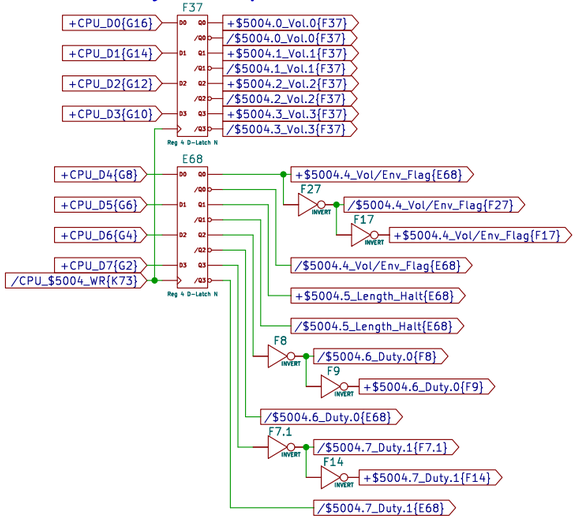
Pulse 1
Duty Cycle Counter



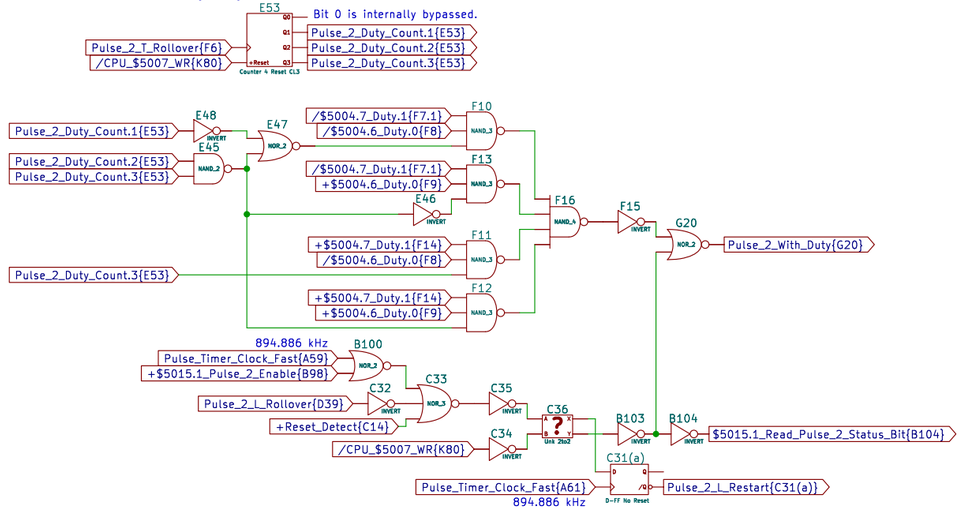
Pulse 1
Envelope/Volume Multiplexing



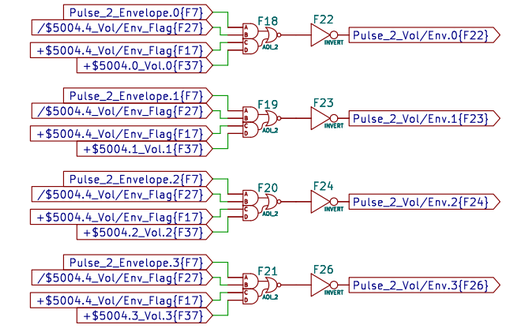
$5004 Register
Pulse 2 Duty Envelope Volume



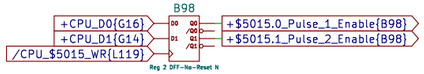
Pulse 2
Duty Cycle Counter



Pulse 2
Envelope/Volume Multiplexing



$5015 Write Register
Pulse 1&2 Status

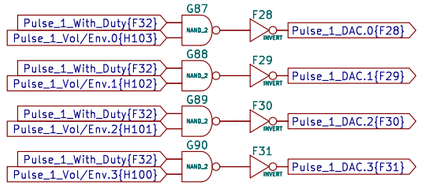


Sheet: /Pulse Duty Envelope Volume/
File: pulse_duty_envelope_volume.kicad_sch

Title:

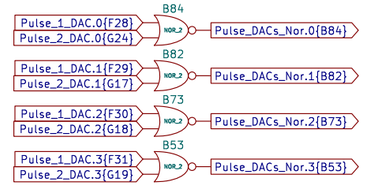
Size: USLedger | Date: | **Rev:**
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Pulse 1&2
Individual 4-bit DAC Outputs

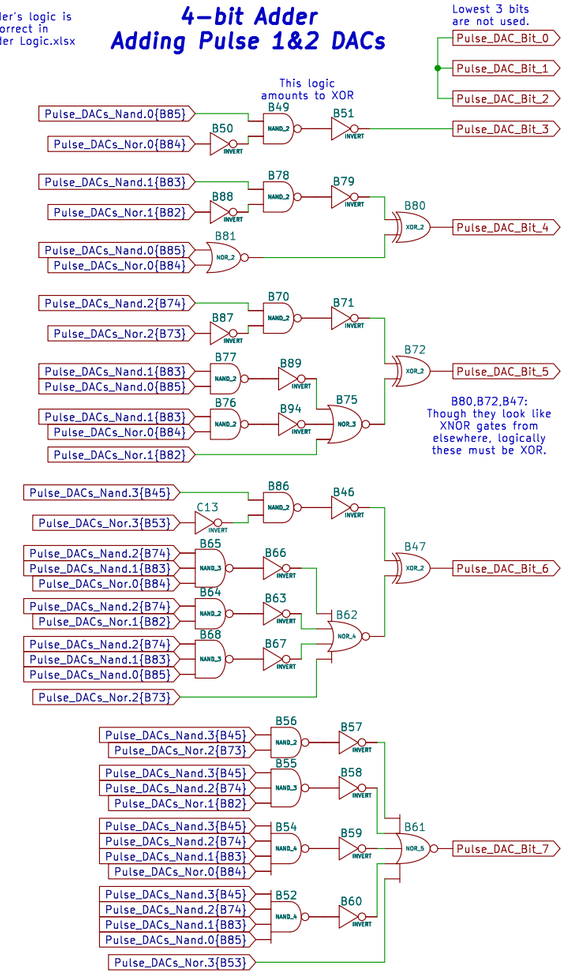


Components for
4-bit Adder

Note: The adder's logic is verified correct in Pulse DAC Adder Logic.xlsx



4-bit Adder
Adding Pulse 1&2 DACs



Sheet: /Pulse DAC/
File: Pulse DAC.kicad_sch

Title:

Size: USLedger | Date: | **Rev:**
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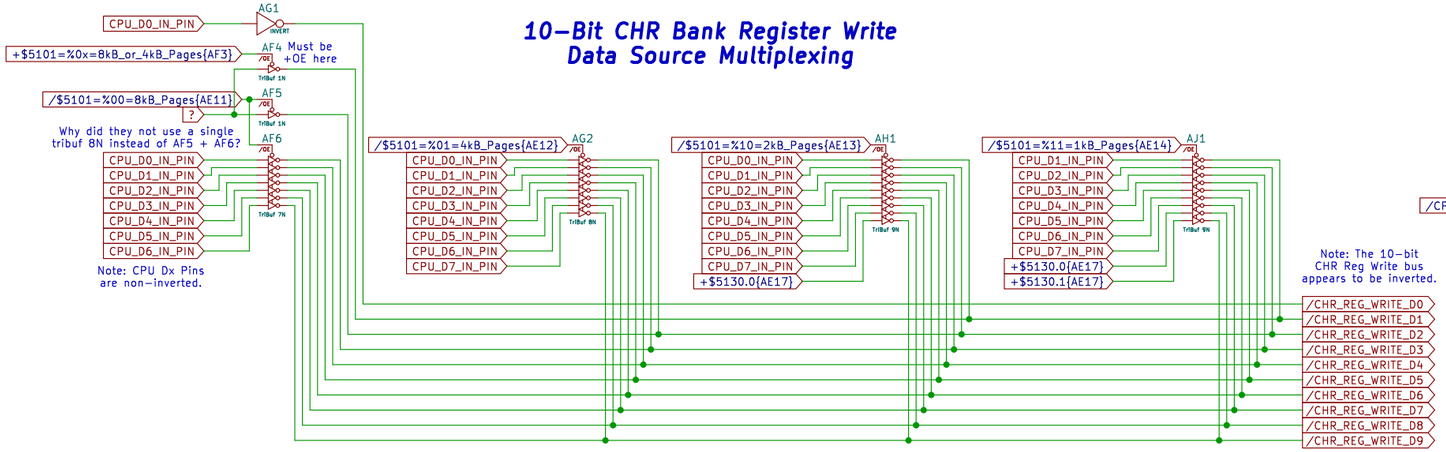
PPU_D0_IN_PIN
PPU_D1_IN_PIN
PPU_D2_IN_PIN
PPU_D3_IN_PIN
PPU_D4_IN_PIN
PPU_D5_IN_PIN
PPU_D6_IN_PIN
PPU_D7_IN_PIN
{BigBuf}
BIGBUF
PPU_D0_In{BigBuf}
PPU_D1_In{BigBuf}
PPU_D2_In{BigBuf}
PPU_D3_In{BigBuf}
PPU_D4_In{BigBuf}
PPU_D5_In{BigBuf}
PPU_D6_In{BigBuf}
PPU_D7_In{BigBuf}
/V-Split_Rendering{BL43}
/Scanline_Counter_Reset{Z25}
/ExRAM_Mode_%01{BL49}
/Extended Attributes Enabled
+Fill_Mode_Nametable_Access{BL33}
/PPU_Attribute_Access{BL25}
+ExRAM_Nametable_Access{A9}
+ExRAM_Mode_$0x{BL48}
BL10
BL34
BL35
BL40
BL32
BL28
BL36
BL37
BL38
BL29
BL30
BL39
BL31
BL41
BL42
BL46
BL45
BK39
BK40
BK42
BK44
BK41
BK43
+$5107.1_Fill_Mode_Color{BL9}
+$5107.0_Fill_Mode_Color{BL8}
BH25
BH27
BH26
+PPU_Fetch_Counter.3{S21}
+V-Split_Scanline_Counter.4{T22}
BH22
BH24
BH20
BH18
BH21
BH23
BH19
BH17
BH4
BH5
BH6
BH7
BH8
BH3
BH2
BH1
ExRAM_D7_Read{BJ12(h)}
ExRAM_D5_Read{BJ12(f)}
ExRAM_D3_Read{BJ12(d)}
ExRAM_D1_Read{BJ12(b)}
+ExRAM_Read_D7_Latch{BJ11}
+ExRAM_Read_D6_Latch{BJ11}
BH12
BH13
BH14
BH15
BH16
BH11
BH10
BH9
ExRAM_D6_Read{BJ12(g)}
ExRAM_D4_Read{BJ12(e)}
ExRAM_D2_Read{BJ12(c)}
ExRAM_D0_Read{BJ12(a)}
ExRAM_D0_Read{BJ12(a)}
+$5106.0_Fill_Mode_Tile{L11}
ExRAM_D1_Read{BJ12(b)}
+$5106.1_Fill_Mode_Tile{L12}
ExRAM_D2_Read{BJ12(c)}
+$5106.2_Fill_Mode_Tile{L14}
ExRAM_D3_Read{BJ12(d)}
+$5106.3_Fill_Mode_Tile{L13}
ExRAM_D4_Read{BJ12(e)}
+$5106.4_Fill_Mode_Tile{L18}
ExRAM_D5_Read{BJ12(f)}
+$5106.5_Fill_Mode_Tile{L17}
ExRAM_D6_Read{BJ12(g)}
+$5106.6_Fill_Mode_Tile{L16}
ExRAM_D7_Read{BJ12(h)}
+$5106.7_Fill_Mode_Tile{L15}
BK1
BL12
BL1
BL13
BL2
BL14
BL3
BL15
BL4
BL16
BL5
BL17
BL6
BL18
BL7
BL19
BK7
BK11
BK15
BK19
BK23
BK27
BK31
BK35
BK10
BK8
BK14
BK12
BK18
BK16
BK22
BK20
BK26
BK24
BK30
BK28
BK34
BK32
BK38
BK36
BK9
BK13
BK17
BK21
BK25
BK29
BK33
BK37
PPU_D0_OUT_PIN
PPU_D1_OUT_PIN
PPU_D2_OUT_PIN
PPU_D3_OUT_PIN
PPU_D4_OUT_PIN
PPU_D5_OUT_PIN
PPU_D6_OUT_PIN
PPU_D7_OUT_PIN
CIRAM_CE_PIN{A9}
+PPU_RD_A{BJ1}
+PPU_A13_B{BK6}
/PPU_A12_B{BK3}
BK2
BK5
BK4
PPU_Dx_Direction{BK4}

Sheet: /PPU Data Bus/
File: ppu_data_bus.kicad_sch

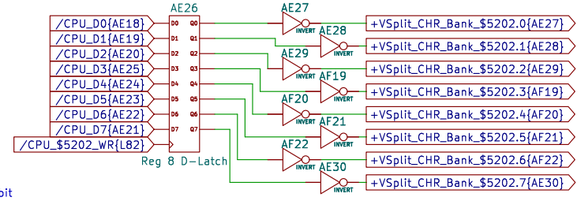
Title:

Size: USLedger | Date: | **Rev:**
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10-Bit CHR Bank Register Write Data Source Multiplexing



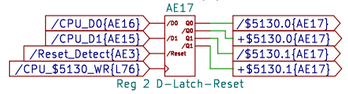
$5202 Register V-Split CHR Bank



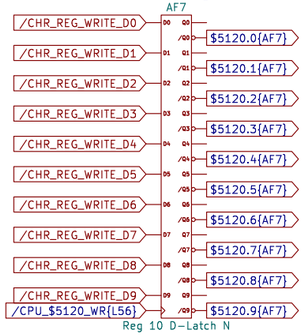
CHR Reg Write Buffering



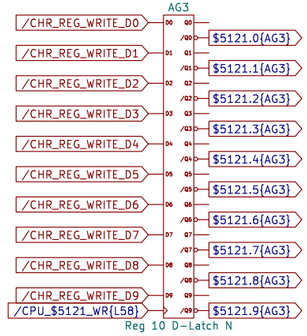
$5130 Register CHR Bank Upper Bits



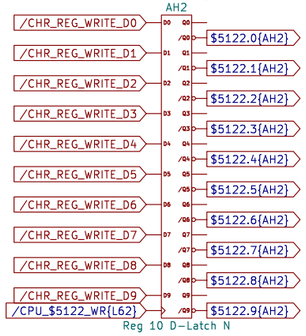
$5120 Register



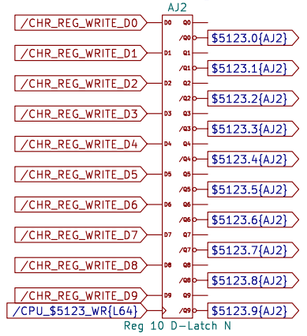
$5121 Register



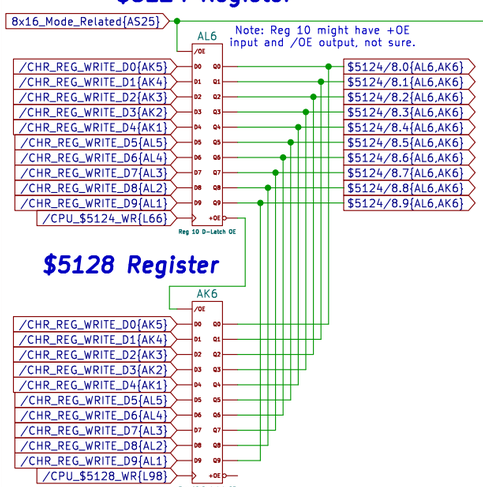
$5122 Register



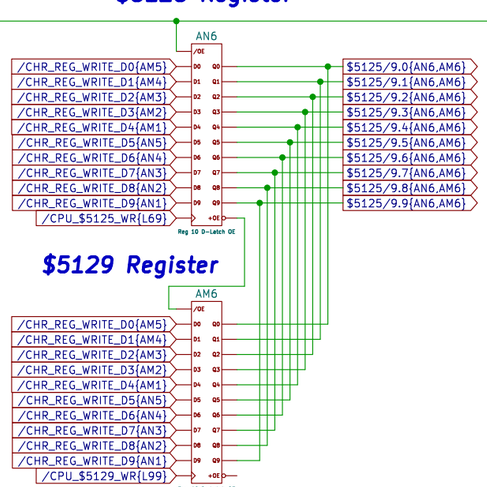
$5123 Register



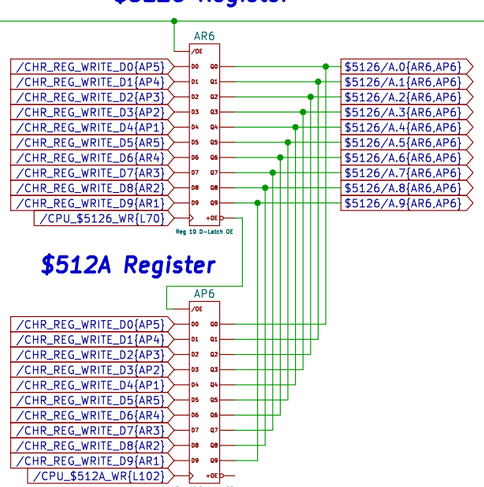
$5124 Register



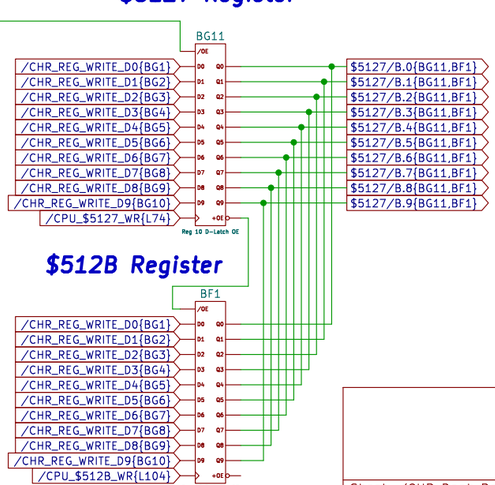
$5125 Register



$5126 Register



$5127 Register



$5128 Register

$5129 Register

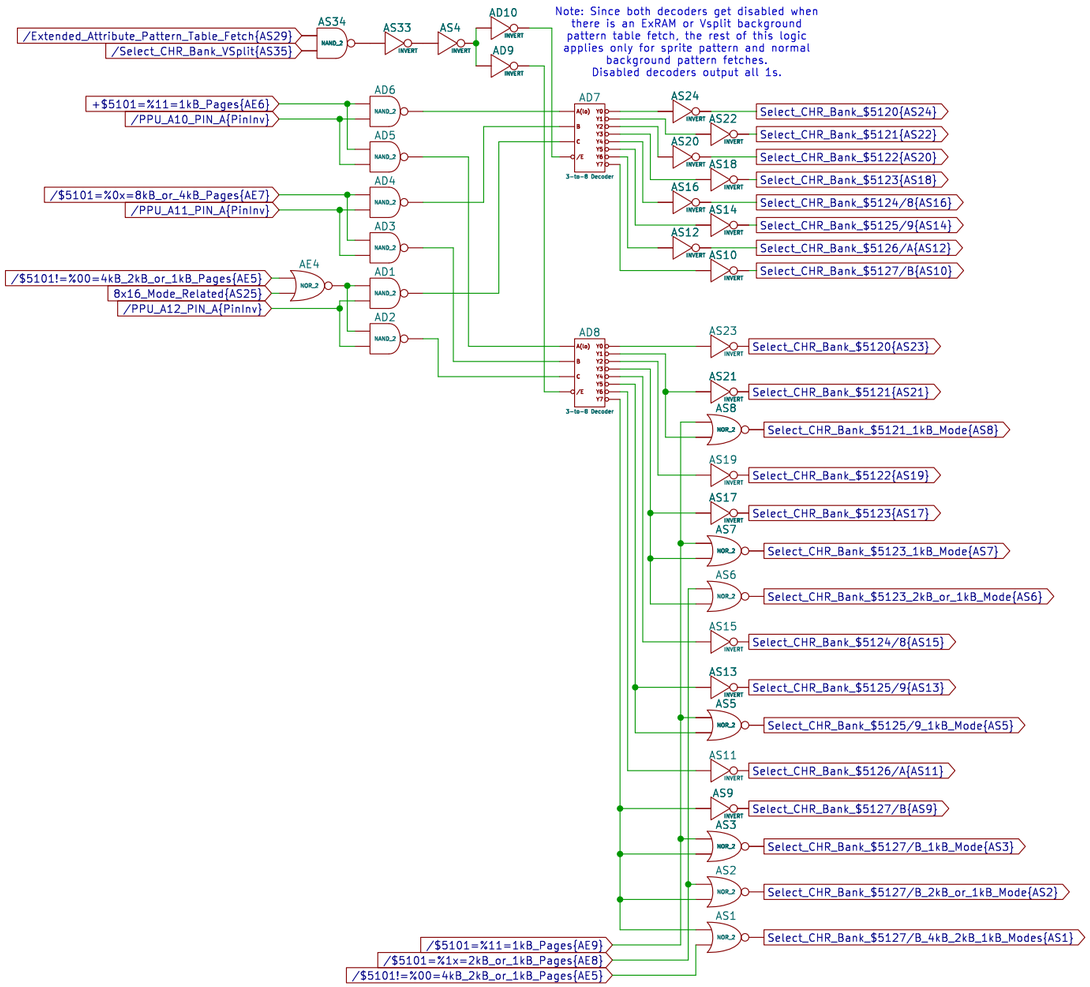
$512A Register

$512B Register

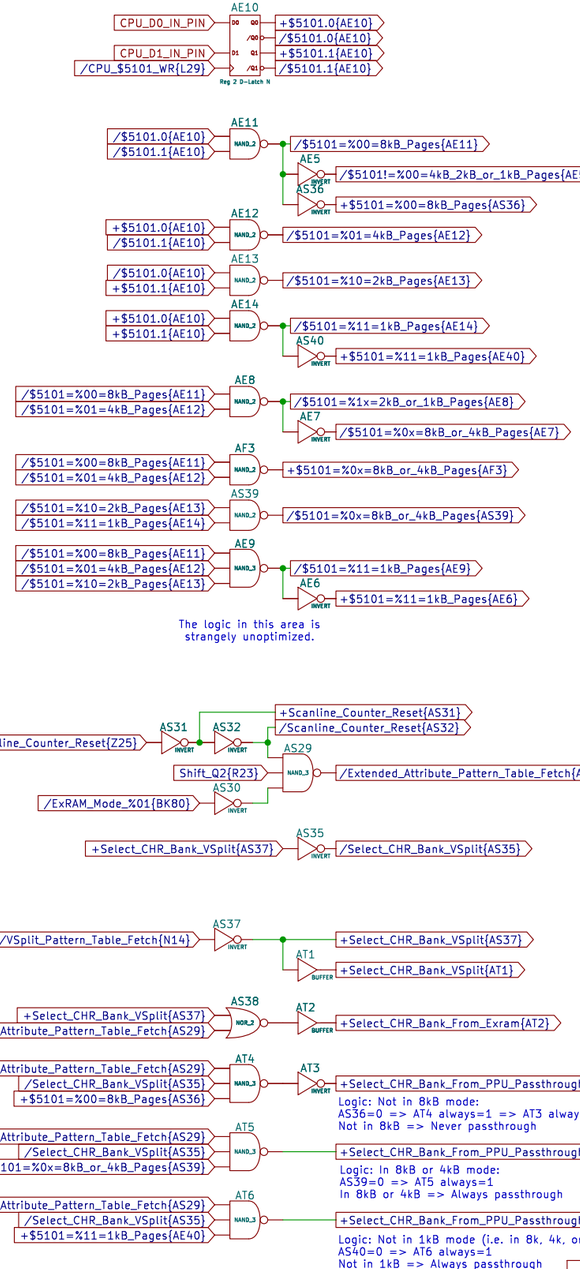
Sheet: /CHR Bank Registers/
File: chr_bank_registers.kicad_sch

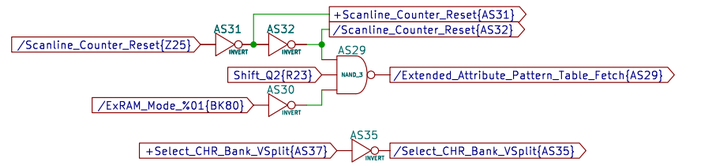
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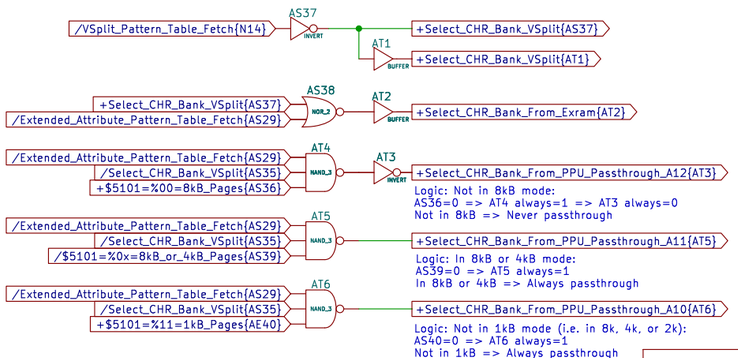
Size: USLedger | Date: | **Rev:**
KiCad E.D.A. 9.0.7 | Id: 21/27



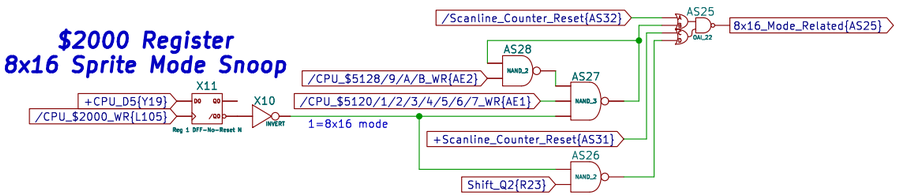
$5101 Register CHR Mode



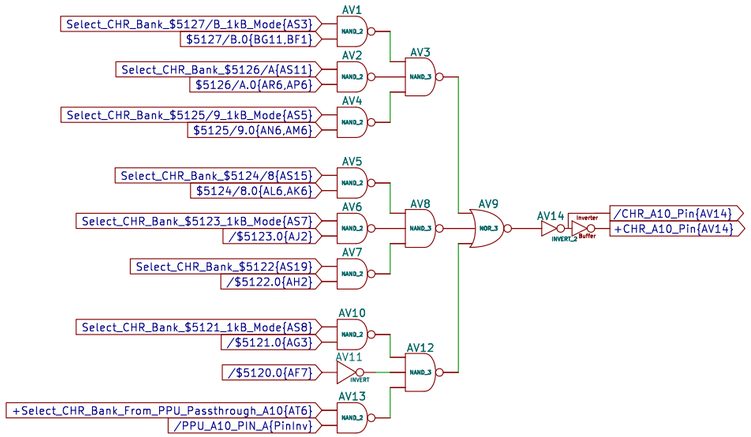




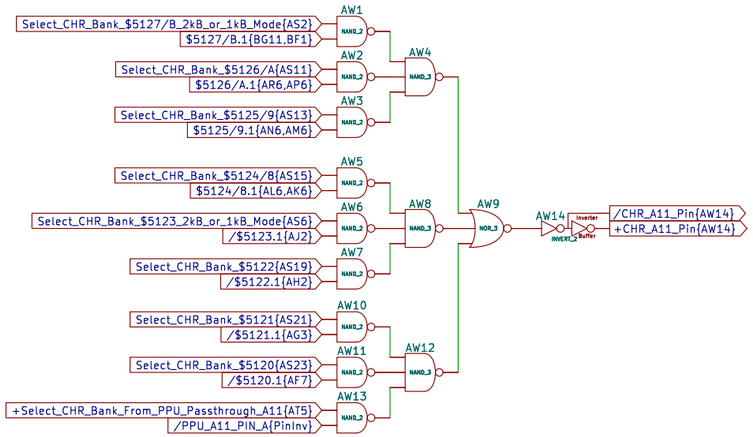
$2000 Register 8x16 Sprite Mode Snoop



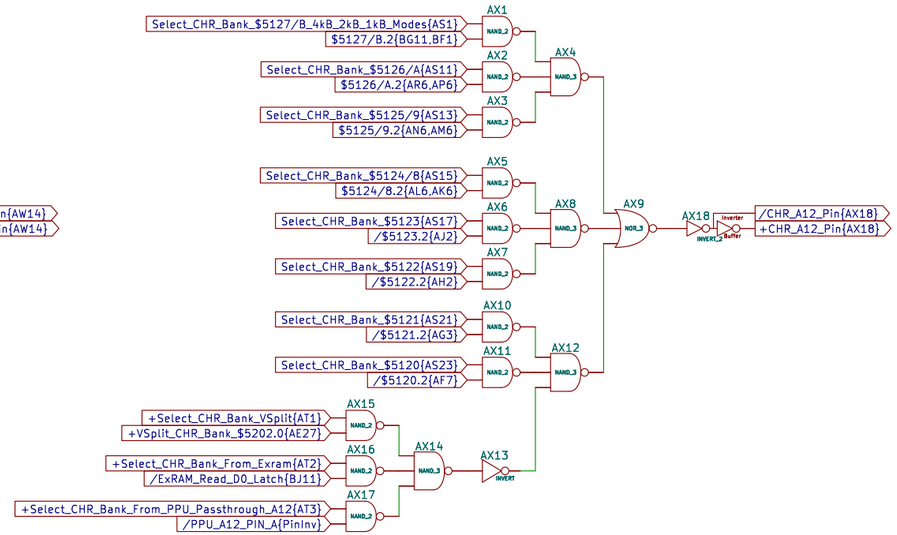
CHR A10



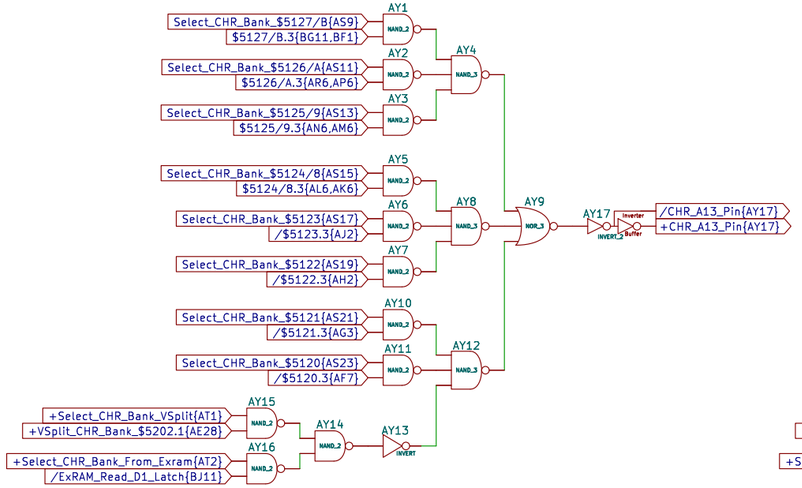
CHR A11



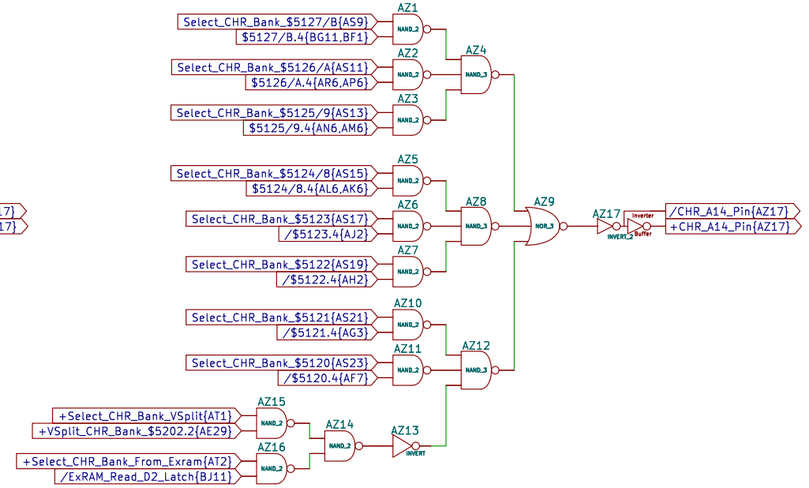
CHR A12



CHR A13



CHR A14

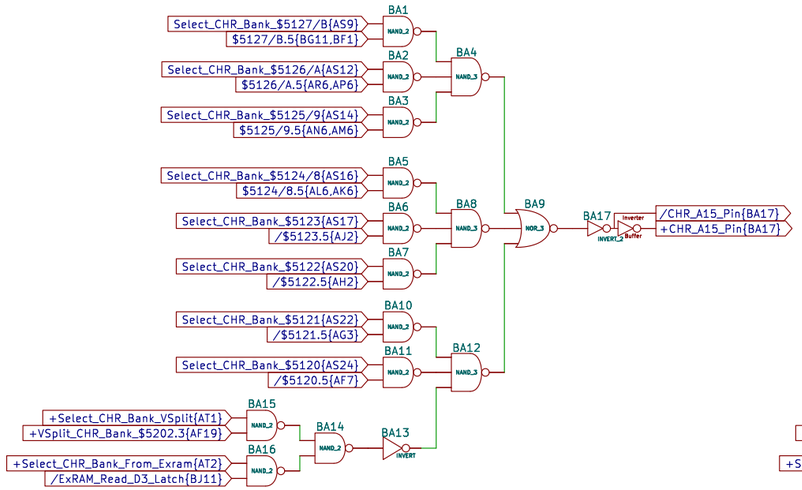


Sheet: /CHR Address Bus 10 to 14/
File: chr_address_bus_10_to_14.kicad_sch

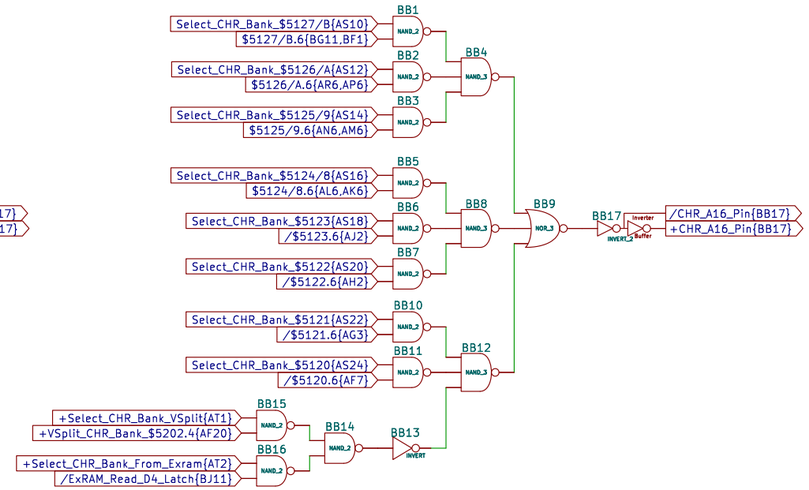
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Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 23/27

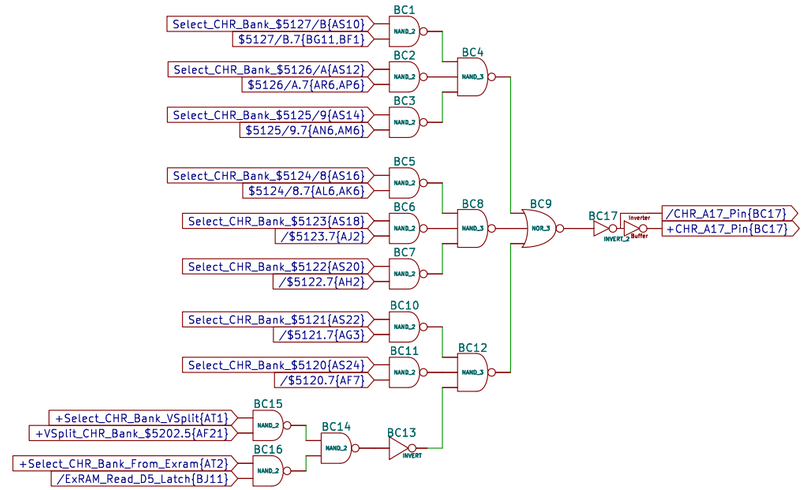
CHR A15



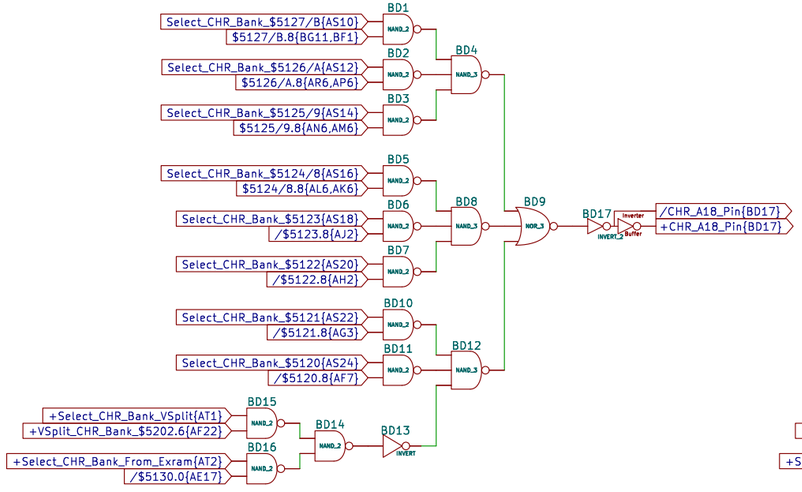
CHR A16



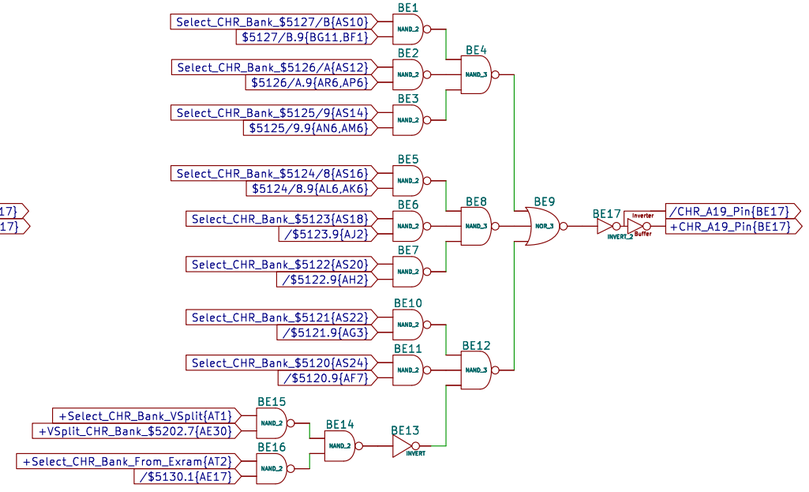
CHR A17



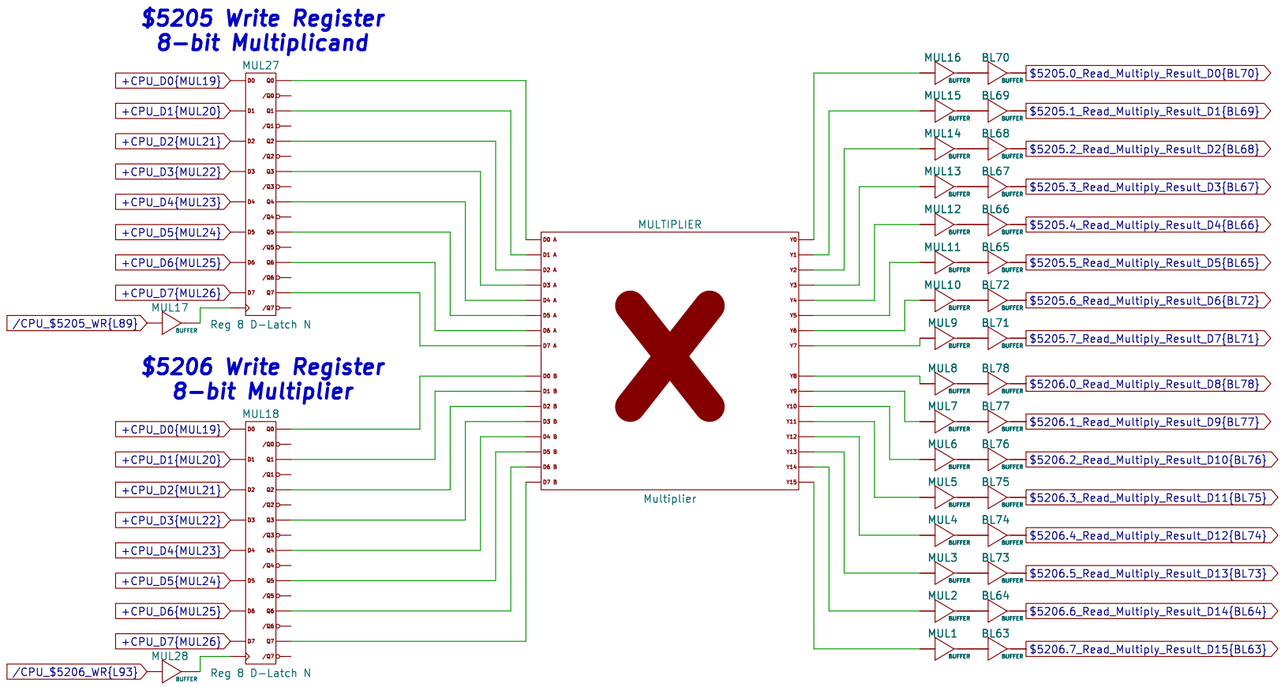
CHR A18



CHR A19



Sheet: /CHR Address Bus 15 to 19/		
File: chr_address_bus_15_to_19.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 24/27

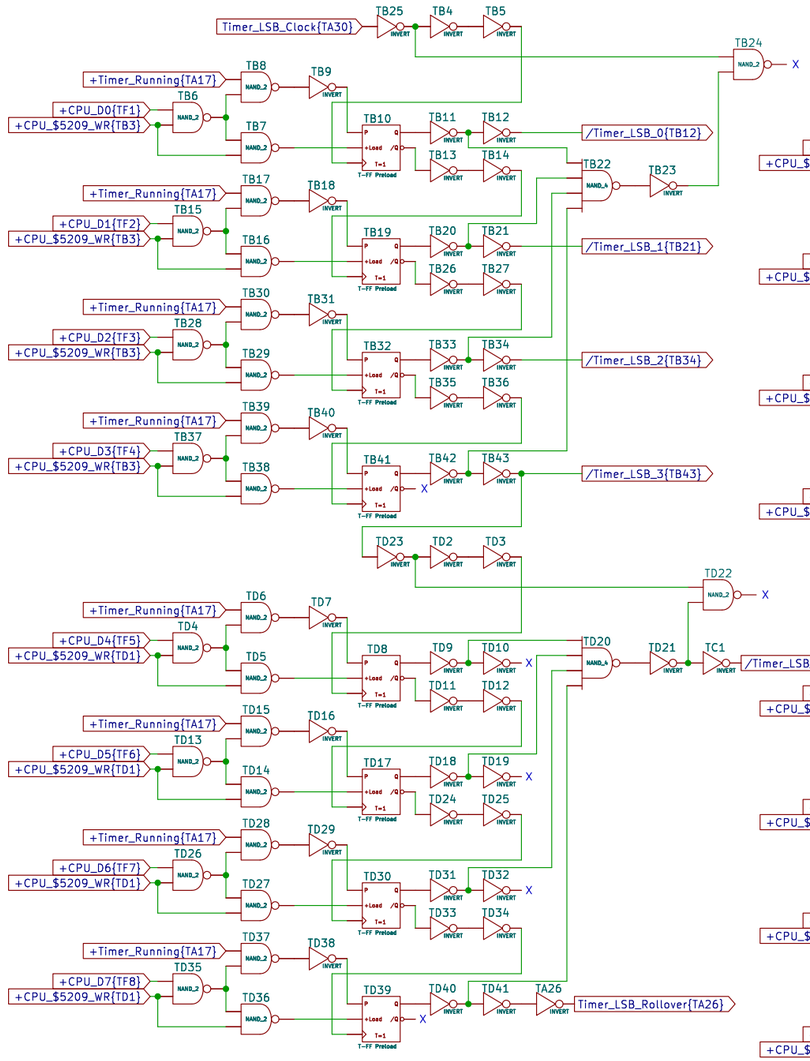


Sheet: /Multiplier/
File: multiplier.kicad_sch

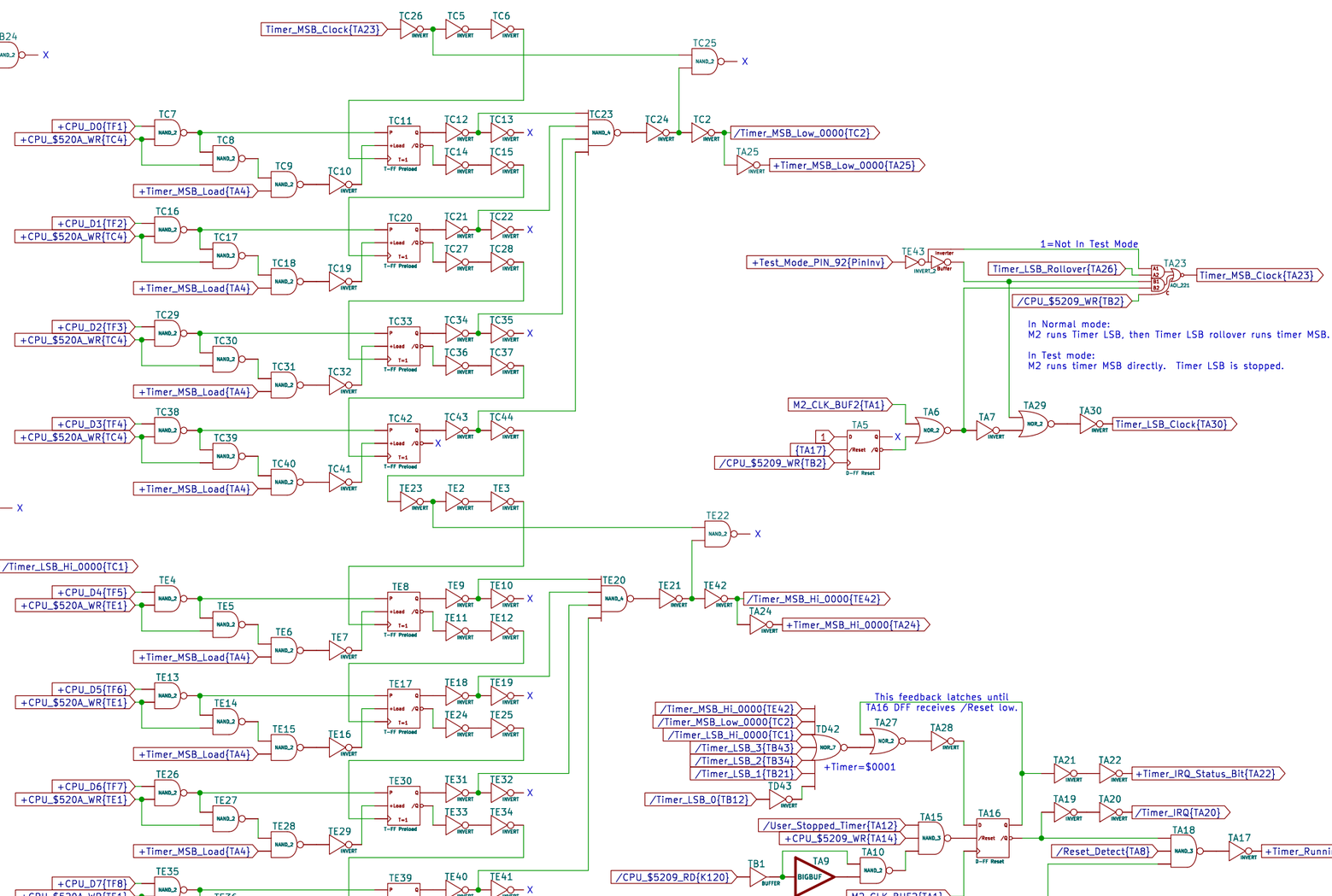
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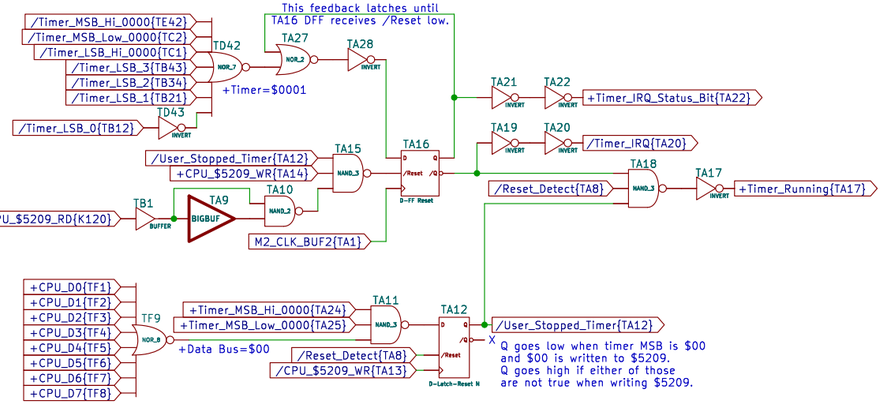
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 25/27

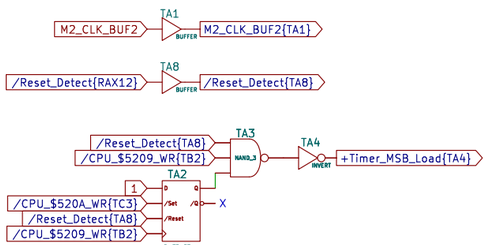
Timer LSB

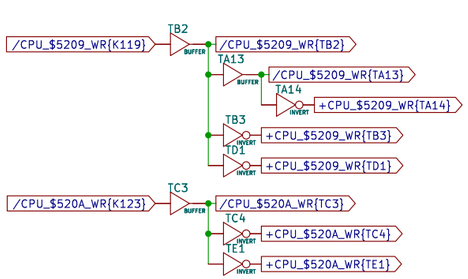


Timer MSB



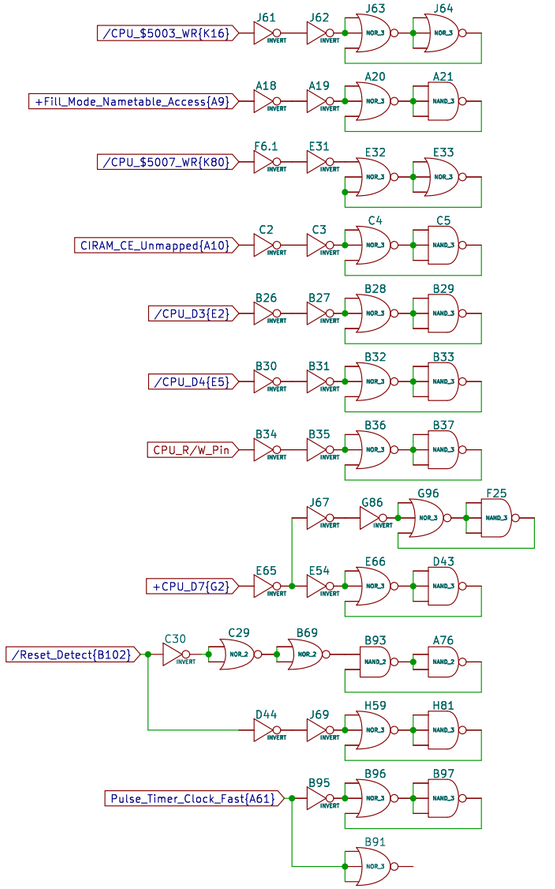




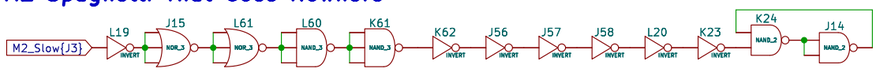


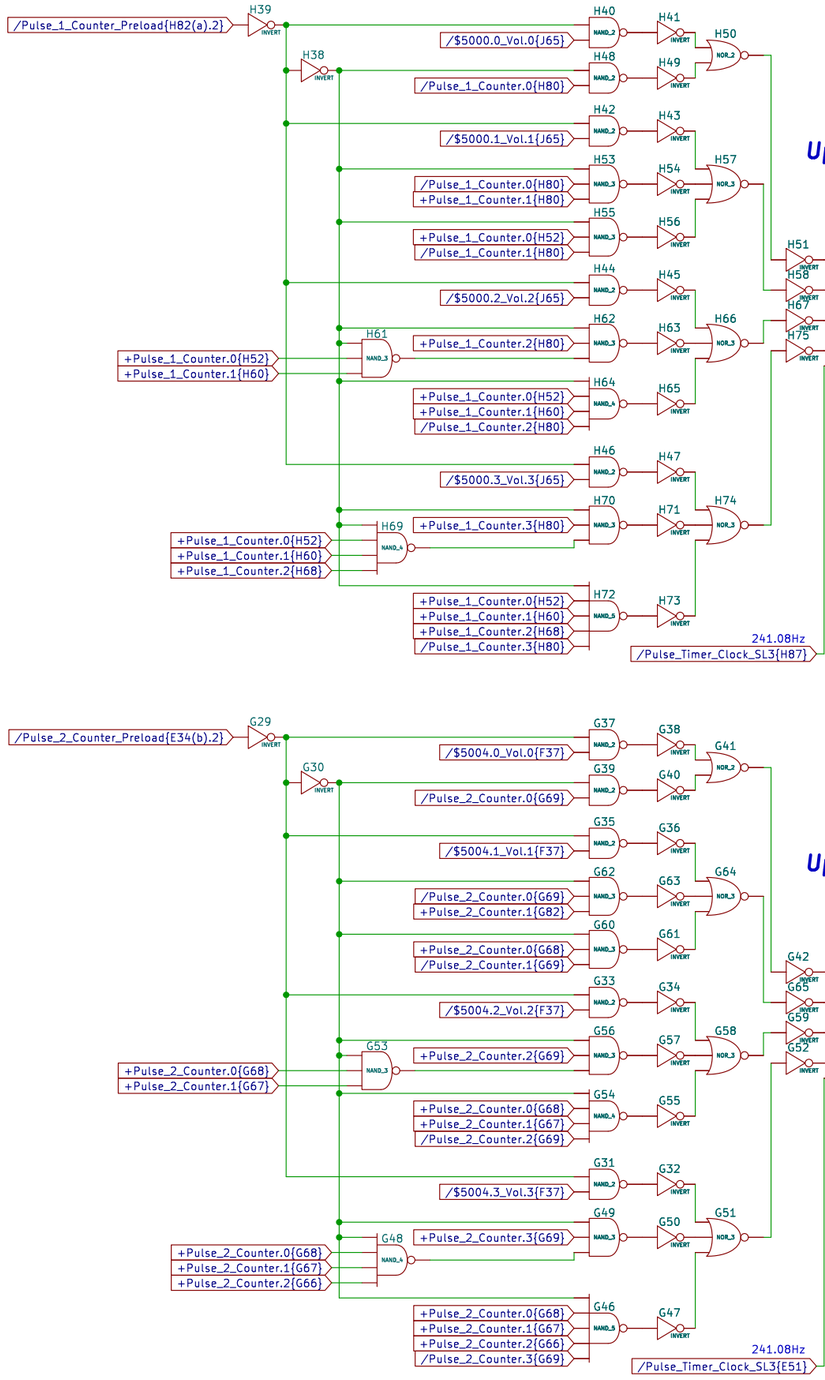
Sheet: /MMC5A Timer/		
File: mmc5a_timer.kicad_sch		
Title:		
Size: USLedger	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 26/27

This sheet contains all residual items that don't serve any apparent purpose.



M2 Spaghetti That Goes Nowhere





Pulse 1 Unused 4-bit Binary Up-Counter With Preload

The only output that goes elsewhere is H78, which gets latched by H82(a), coming directly back here as /Pulse_1_Counter_Preload; all self-contained. It does not appear to provide any other external signals, so I don't think it actually does anything. It seems like an abandoned envelope counter that was left in.

It latches the 4-bit pulse volume, then counts up at a rate of 240 Hz. Nothing stops it from counting or rolling over from $F back to $0.

Pulse 2 Unused 4-bit Binary Up-Counter With Preload

Sheet: /Artifacts/	
File: artifacts.kicad_sch	
Title:	
Size: USLedger	Date:
KiCad E.D.A. 9.0.7	Id: 27/27